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ImageSense: Caption & Retrieve Mastery

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Abstract

Image captioning and Content-based image retrieval are important tasks for improving human-computer interaction as well as for a deeper understanding of the visual content. These domains have been a dynamic and rapidly progressing area of research since the 1990s. Over the last ten years, significant advancements have been achieved in both theoretical exploration and the creation of systems. However, even the best models have a limit in the quality of the results with a lack of training data. In order to address this and introduce diversity into our dataset, our project will use a novel data augmentation approach using text-to-text and text-to-image generative models [1]. This technique is used to create high-quality augmented datasets, especially useful for robust image captioning and CBIR in data-constrained situations. Augmenting both text and image modalities, rather than focusing solely on one, brings significant benefits.

Executive Summary

This document is about our development-based final year project, named ImageSense: Caption & Retrieve Mastery. This report's objective is to familiarize with the features, intended audience and attributes of ImageSense. The problem identified was that Journalists often work with images to support their stories. Without automated image captioning, journalists would need to manually write descriptions for each image they use in their articles and if they want to search information about an image they would have to rely on metadata or keywords to search for images in their databases, which might not accurately reflect the content of the images. This can make the process more time consuming and less accurate. To solve these problems, we will develop two key solutions: automated image captioning and content-Based Image Retrieval software. Automated image captioning involve generating accurate and contextually relevant descriptions for images and CBIR software allows journalists to search for images based on their content, visual attributes and features rather than relying solely on metadata or keywords.

Our project's goals and objectives are focused on developing comprehensive software solution that utilizes data augmentation technique to train models using large and diverse datasets. This attempt seeks to boost the effectiveness, precision, and quality of journalists tasks by providing automated image captioning and CBIR solutions. In the end, our goal is to empower them to create captivating visual stories and simplify their image related tasks.

The project scope which we identified is that we will be developing a fully functional website that will be composed of two main modules. The first module will allow users for generating accurate and contextually relevant descriptions for their images. The second module enables users to develop a CBIR system, empowering journalists to search for images using visual characteristics like color, shape, and composition. We also identified the stakeholders for our project, which included journalists, photo editors and graphic designers.

We have also discussed our literature survey and related work regarding image augmentation, image captioning and image retrieval. We explored NLPAug; a commonly used framework that is used for text to text augmentation and two text to image augmentation techniques that are Generative Adversarial Networks (GANs) and Stable Diffusion. For image captioning, LSTM and Bi-LSTM are considered as the conventional and fundamental approaches. However, in more recent years, Transformer models like those proposed by Kumar have become the new cutting-edge techniques. For CBIR, we examined the research papers of Ray-I Chang and Ponitz.

Software requirements specifications are also mentioned in this project, which also includes a list of features. Some of the major features included are image captioning, image retrieval and many other

features. We have also enlisted the functional requirements for users. Along with this, we have mentioned non-functional requirements. This section also provides a list of prominent use cases such as sign up, image uploading and save output e.t.c. Moreover, a simple GUI mockup of our website is also provided, which includes the home page, features, and other main pages. The document also outlines certain alleged risks that might hinder our development process or affect how our application works for users.

Our document comprehensively addresses both the system's high-level and low-level components, encompassing its design, particularly focusing on the system architecture. This architecture is divided into two distinctive sub-architectures, specifically tailored for image captioning and image retrieval. We have also enlisted the architectural strategies, offering a strategic perspective. The document presents a series of essential sequence diagrams like login, signup, image upload, and download. Additionally, a comprehensive class diagram is included for a comprehensive understanding. In the last section, we conclude by summarizing policies and tactics that wield influence over the system's interface and implementation, providing a comprehensive overview of our design approach and considerations.

Finally, to end the document, we have provided a comprehensive overview of the implementation undertaken thus far.

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Chapter 1 Introduction

In a world increasingly driven by visual content and rapid technological advancements, the fields of image captioning and CBIR have emerged as crucial areas. This chapter provides a brief introduction which focuses on augmenting data to enhance the performance of image captioning and CBIR systems. With the expansion of massive image datasets and the rise of transformer-based models, the need to ensure optimal performance in resource-constrained scenarios becomes primary. By utilizing innovative text-to-text and text-to-image generative methods, this project seeks to address the challenges presented by limited data availability, and ultimately will enrich the quality of image captioning and enabling more efficient image retrieval.

1.1 Purpose of this Document

This project report aims to present a thorough strategy for improving image captioning and content-based image retrieval through innovative data augmentation techniques. By utilizing text-to-text and text-to-image generative models to create diverse variations of images and captions the project aims to address the difficulties caused by limited data availability in these domains. By doing so, it seeks to elevate the performance of models in scenarios with constrained resources. It will ultimately enable users to comprehend images more comprehensively and facilitating efficient reverse image searches. The project's scope covers the exploration and development of image captioning models and CBIR systems. The document also touches preliminary studies conducted in the field, highlighting the significance of transformer models, traditional approaches, and CBIR methods.

1.2 Intended Audience

Following will be the intended Audience:

- The Development Team will use this document to understand the requirements so that they can implement and develop the application accordingly.
- Users that includes journalists, photo editors and graphic designers will utilize this document to understand the functionality and details regarding the use of the system.
- Supervisor will use the document to identify the development needs.
- Panel of professors will use this document while evaluating the project.

1.3 Definitions, Acronyms, and Abbreviations

The key acronyms used in this document are listed below:

CBIR: Content-Based Image Retrieval

GAN: Generative Adversarial Networks

LSTM: Long Short-Term Memory

Bi-LSTM: Bidirectional Long Short-Term Memory networks

GUI: Graphical User Interface

SDG: Sustainable Development Goal

SQL: Structured Query Language

VRAM: Video Random Access Memory

GPU: Graphics Processing Units

CSS: Cascading Style Sheets

VScode: Visual Studio Code

ERD: Entity-Relationship Diagram

PK: Primary Key

FK: Foreign Key

AI: Artificial Intelligence

1.4 At the End of Introduction Chapter

The document provides a clear and detailed description of the project's vision, outlining its main goals and objectives. It then highlights the pertinent work concerning enhancing text and images, generating image descriptions, and Content-Based Image Retrieval. Following this, it delineates all the specific requirements of the project. This document acts as a formal agreement among stakeholders, facilitating developers in comprehending and implementing features that align with stakeholder requirements and usability.

Chapter 2 Project Vision

The following section describes the problem statement, scope and objectives of this project while also explaining the business opportunity and list of stakeholders.

2.1 Problem Domain Overview

This project focuses on image captioning and content-based image retrieval, ensuring data scarcity, caption quality, and efficient image retrieval. It aims to enhance model performance in resource-constrained scenarios by employing advanced data augmentation techniques, creating diverse datasets for improved image understanding and reverse image searches. Professionals like journalists will benefit from automated image captioning and CBIR tools, streamlining tasks and providing contextually relevant image descriptions, reducing reliance on metadata, and enhancing productivity in handling visual content.

2.2 Problem Statement

A detailed problem statement is shown in 2.1.

Table 2.1: Problem Statement

The table describes the problem statement, the effect and solution to it

Problem	Journalists often use images to enhance their articles, but the lack of automated image captions means they have to write descriptions manually, which is time-consuming. Moreover, finding specific image details relies heavily on metadata and keywords, which can lead to content inaccuracies in databases.
Effect	Spending a lot of time creating descriptions for images and relying heavily on metadata and keywords for image retrieval.
Solution	A platform that allows users automated image captions and the development of CBIR software to search for images based on their content, visual attributes, and features rather than relying solely on metadata or keywords.

2.3 Problem Elaboration

The problem at the core of this project is the significant burden placed on journalists due to the absence of automated image captioning tools and image retrieval tools. Journalists regularly incorporate images into their articles to enhance visual storytelling but the manual task of composing descriptive captions for each image is a laborious and time consuming attempt. Additionally, reliance on metadata and keywords for retrieving specific image details within databases introduces a risk of inaccuracies and content inefficiencies. The adverse effects ripple through the journalism landscape, compromising efficiency, accuracy, and the overall quality and accessibility of journalistic content. To mitigate these challenges and their impacts, the proposed solution seeks to develop a comprehensive platform that not only automates image captioning but also pioneers the advancement of Content-Based Image Retrieval software. This approach aims to empower journalists with efficient tools for creating captivating visual narratives and will also address the challenges posed by manual tasks and metadata dependent image retrieval processes.

2.4 Goals and Objectives

Our project goals are as follows:

- Develop a fully functional ImageBot.
- Utilize data augmentation techniques.
- Train models with large and diverse datasets.
- Boost effectiveness, precision and quality of journalists' tasks.
- Provide automated image captioning and CBIR solutions.
- Empower journalists to create captivating visual stories.
- Simplify image related tasks.

Our objectives behind this project are to:

- Research and select appropriate data augmentation methods.
- Preprocess diverse datasets for training.
- Explore machine learning models for image captioning and CBIR.
- Integrate the software solution into journalists workflows.
- Build a robust and scalable software infrastructure.

- Ensure user friendly interfaces for ease of use.

2.5 Project Scope

The project's scope encompass the development of a comprehensive software solution, known as "ImageSense: Caption and Retrieve Mastery." The aim is to revolutionize image handling and interpretation by addressing the challenges associated with image captioning and content-based image retrieval. The project will consist of two primary modules:

The first module focuses on automated image captioning leveraging advanced data augmentation techniques like text-to-text and text-to-image augmentation. The goal is to generate accurate and contextually relevant descriptions for images. By enhancing the quality of image captions this module aims to streamline the process of image description, reducing manual effort and improving precision.

The second module is content-based Image Retrieval. It will empower users to search for images relying on visual characteristics like color, arrangement and form rather than relying solely on metadata or keywords. This module will utilize augmented image data to enhance retrieval accuracy and efficiency. It will also allow users to find images that match their specific visual criteria. Both modules will be integrated into a user friendly website, ensuring accessibility and usability for a diverse range of users including journalists, students and social media enthusiasts. The success of this project will be measured by its ability to deliver a robust and efficient software solution that fulfills the identified goals and objectives and also addresses potential risks and challenges

2.6 Sustainable Development Goal (SDG)

Sustainable Development Goal of this project focuses on industry, innovation, and infrastructure. This project leverages image processing techniques to drive innovation in the fields of computer vision and natural language processing, aiming to enhance image understanding and retrieval processes for various industries. It promotes efficient use of visual content by journalists, students, and social media users. The initiative aligns with SDGs, emphasizing sustainable development and improved storytelling for better global issue comprehension.

2.7 Constraints

The following are the constraints to the system:

- There is a limit on the number of pictures that users can upload at a time for image captioning.

- The models will not perform well on low quality images.
- Models can be biased because of the biased data on which they are trained on.
- Models can be computationally expensive to train and deploy.

2.8 Business Opportunity

The fusion of image augmentation, image captioning and content-based image retrieval in a software platform presents an exciting business opportunity for journalists and media professionals. In today's fast paced digital media landscape, visuals are key to capturing and retaining audience attention. This software can empower journalists to enhance their storytelling by augmenting images making them more compelling and aligned with the narrative. Additionally the image captioning feature adds context and depth to images that aids in creating engaging captions that resonate with the audience. CBIR facilitates effective archiving and retrieval of images that streamlines research and enhancing the overall journalistic workflow. Journalists can leverage this technology to enrich their articles and to create visually appealing content and to efficiently manage their media assets that ultimately leads to more engaging and informative storytelling.

2.9 Stakeholders Description/ User Characteristics

The primary stakeholders and users of this software are as follows:

- Journalists and Media Professionals
- Photo Editors and Graphic Designers
- Editorial Managers and Newsroom Decision-makers

2.9.1 Stakeholders Summary

The primary stakeholders and users of this software are journalists and media professionals. Journalists are the main users equipped with an understanding of the power of compelling visuals in their work. They seek to enhance their storytelling and engage their audience by leveraging augmented images and creating meaningful captions. Photo editors and graphic designers are also experts in image curation and design focusing on selecting and optimizing visuals to align with the narrative. They require an intuitive interface and efficient tools for image editing and augmentation. Editorial managers and newsroom decision makers are additional stakeholders, emphasizing streamlined workflows and resource management. They look for software that enhances productivity and ensures high quality journalistic output. Collab-

oration and feedback loops among these stakeholders are critical to tailor the software to the specific needs and expectations of the journalism industry.

2.9.2 Key High-Level Goals and Problems of Stakeholders

The stakeholders seek to achieve high level goals such as improving storytelling by leveraging augmented images, enhancing audience engagement through compelling visuals and captions, optimizing image curation and design processes, streamlining workflows and ensuring resource efficiency. Key problems to address include the need for an intuitive interface, efficient image editing and augmentation tools, effective collaboration and feedback mechanisms, and alignment with journalistic requirements for impactful and high-quality output.

2.10 Conclusion

In a nutshell, this chapter offers a thorough understanding of the problem domain. This innovative approach not only addresses the constraints and business opportunities of the domain but also contributes to the comprehensive SDGs of promoting journalistic excellence.

Chapter 3 Literature Review / Related Work

This section provides a comprehensive survey and synthesis of existing knowledge, theories, and research relevant to the image augmentation, captioning and retrieval.

3.1 Definitions, Acronyms, and Abbreviations

All significant abbreviations utilized within this document are as follows:

BERT: Bidirectional Encoder Representations from Transformers

LM: Language Model

CLIP: Contrastive Language-Image Pretraining

ViT: Vision Transformer

UNET: U-Network

DAMSM: Deep Attentional Multimodal Similarity Model

CNN: Convolutional Neural Network

RNN: Recurrent Neural Networks

GRU: Gated Recurrent Unit

BLEU: Bilingual Evaluation Understudy

SIFT: Scale-Invariant Feature Transform

SURF: Speeded Up Robust Features

3-D: 3-Dimensional

IGA: Interactive Genetic Algorithm

BoVW: Bag of Visual Words

3.2 Detailed Literature Review

Due to technological advancements, considerable advancements have been achieved in the realm of images. We extensively reviewed research papers related to our area of study to acquire knowledge about prior work related to data augmentation, captioning, and image retrieval tasks. In the last ten years, significant strides have been achieved in both theoretical research and the development of systems.

3.2.1 Data Augmentation

In the realm of computer vision, augmentation has been extensively employed to enhance the generalization and resilience of machine learning models. Its widespread utilization stems from its ability to artificially expand the training dataset, providing diverse and representative examples for the model to

learn from.

Several techniques for data augmentation are discussed:

3.2.1.1 Text-to-Text Augmentation:

Atliha and Sesok [2] used five diverse text augmentation techniques:

1. Word Deletion - Transforms an input by removing a randomly selected word
2. Synonym replacement - Transforms an input by replacing its words with synonyms provided by WordNet
3. Embedding swap - Transforms an input by replacing its words with similar words from embedding space
4. Inner swap random - Transforms an input by reordering the words in the sentence
5. BERT-Masked LM swap - Generate potential replacements for a word using a masked language model.

The robustness in this methodology is that they used the popular NLPAug framework for implementations of these attacks. For any input sentence, they restricted their model to only modify one word. The portion of the input sentence to be modified is a hyperparameter which in practice is set based on the length of each input. The weakness is potential loss of context, over-reliance on language models, and the challenge of generating truly diverse and coherent captions, which may affect the quality and relevance of augmented data. Text-to-Text augmentation is relevant for image captioning as it enriches the caption dataset by generating diverse variations, improving model generalization, robustness, and the ability to describe images more creatively and accurately.

3.2.1.2 Text-to-Image Augmentation:

In the context of image augmentation, several generative models have demonstrated the ability to generate new images based on input prompts that are as follows:

- **Stable Diffusion Model**
- **Generative Adversarial Networks**

Stable Diffusion Model:

According to Robin Rombach [3], It's a Latent Diffusion Model employing a static, In the context of imagetext encoder pretrained (CLIP ViT-L/14). The components of a stable diffusion model as shown in Figure 3.1 include a a text encoder, image feature extractor, and image decoder (the last two together

can be referred as an image generator).

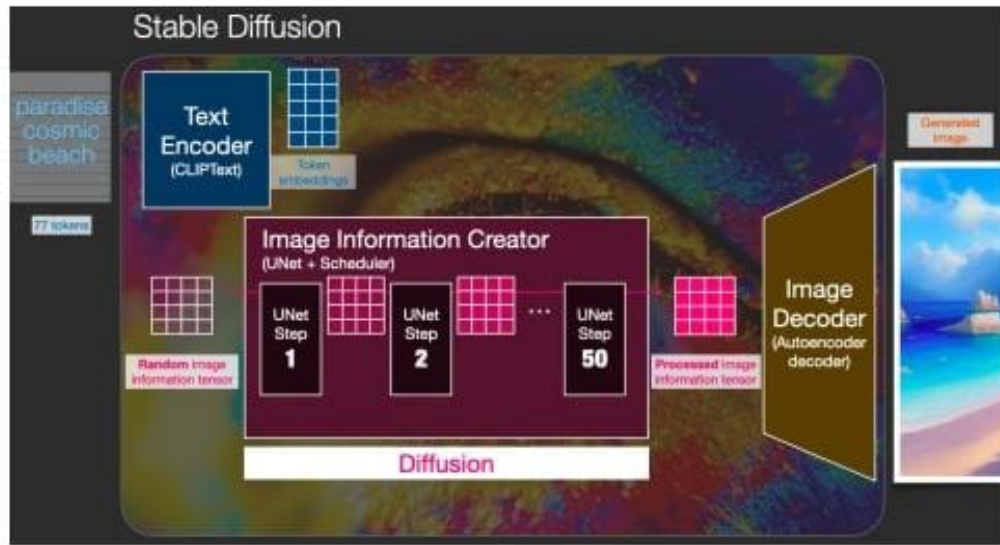


Figure 3.1: Steps in a stable diffusion model (with sample prompt and generated image)

This figure represents the components of a stable diffusion model that includes a text encoder, image feature generator, and image decoder

- **Text encoder:** This component, responsible for text understanding, translates textual information into a numeric representation that encapsulates the ideas conveyed in the input prompt. The stable diffusion model makes use of a special Transformer language model - CLIP. It receives the input text and generates a list of numeric values, where each number corresponds to a word or token in the text (a vector per token). This data is subsequently fed to the Image Generator.
- **Image information creator:** This element serves as the distinctive element of Stable Diffusion. The image information generator operates exclusively within the image information space, also known as the latent space. This characteristic enhances its speed compared to earlier diffusion models operating in pixel space. Comprising a UNet architecture and a scheduling mechanism, this component embodies the concept of "diffusion," representing the incremental processing of information culminating in the creation of a high-quality image.
- **Image Decoder:** The image decoder interprets the data received from the information creator into a picture. It executes only once at the end of the process to produce the ultimate pixel image.

The model's resilience is achieved through a gradual and controlled alteration of text embeddings, allowing for seamless transitions within the generated image space. This process preserves both the semantic meaning and diversity of the resulting images. On the other hand, the model may include challenges in capturing complex image features and potential constraints in generating highly detailed or specific images from text embeddings. Our approach will use this technique to facilitate the controlled and co-

herent generation of diverse images corresponding to text embeddings, enhancing the quality and variety of augmented data for image-related tasks.

Generative Adversarial Networks:

As stated by Tao Xu and Pengchuan Zhang [4], In models for generating images from text based on GANs, the usual approach involves encoding condensing the entire sentence text description into a singular vector for image generation conditions. However, these models lack detailed information at the word level. [5].

Their emphasis was on the **Generative Network**, which draws distinct subregions of the image using words that are especially pertinent to those subsections. The suggested attentional generative network comprises m generators: $G_0, G_1, \dots, G_{m-1}$. These generators take the hidden states $(h_0, h_1, \dots, h_{m-1})$ as input and produce images of varying scales, from small to large $(\hat{x}_0, \hat{x}_1, \dots, \hat{x}_{m-1})$. Secondly, they calculated the detailed image-text alignment loss using DAMSM.

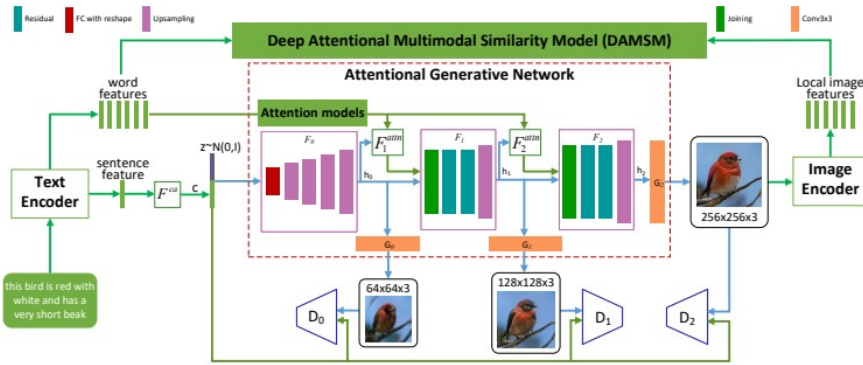


Figure 3.2: The structure of the suggested AttnGAN

This figure represents the components of AttnGAN model that includes a text encoder, attention model and image encoder

The **DAMSM**(Deep Attentional Multimodal Similarity Model) entails training two neural networks as shown in Figure 3.2. These networks map subregions aligning the image and sentence words into a shared semantic space. This enables the model to evaluate image-text similarity at the word level, facilitating the calculation of a detailed loss during image creation.

- **Text Encoder** A bidirectional Long Short-Term Memory (LSTM) [6] operates as the textual encoder, extracting semantic vectors from the textual description. In this bidirectional LSTM, each word possesses two corresponding hidden states, one for each direction. Consequently, the concatenation of these the combination of these two hidden states is utilized to represent the semantic significance of a word.
- **Image Encoder:** The Image Encoder is a Convolutional Neural Network (CNN) engineered to

convert images into semantic vectors. In this CNN, the intermediate layers are dedicated to grasping local features from different parts of the image, while the subsequent layers concentrate on capturing overall features of the entire image. Initially, the input image is resized to dimensions of 299×299 pixels. Subsequently, they extracted the reshaped local feature matrix $f \in \mathbb{R}^{768 \times 289}$ obtained from the "mixed 6e" layer of Inception-v3, (initially with $768 \times 17 \times 17$) denoted as f is obtained. Each column in f denotes the feature vector of a particular subsection of the image. The local feature vector has a dimension of 768, corresponding to the 289 sub-regions present in the image. Simultaneously, the global feature vector $\bar{f} \in \mathbb{R}^{2048}$ is extracted from the final average pooling layer of Inception-v3. To achieve alignment in a common semantic space with text features, a perceptron layer is applied to convert the image features.

$$\mathbf{v} = \mathbf{W}\mathbf{f}, \bar{\mathbf{v}} = \bar{\mathbf{W}} \cdot \bar{\mathbf{f}} \quad (3.1)$$

where $\mathbf{v} \in \mathbb{R}^{D \times 289}$ and its i^{th} column e_i represents the visual feature vector i^{th} for a sub-region of the image; and $\bar{\mathbf{v}} \in \mathbb{R}^D$ represents the global vector for the entire image.

GANs has the power to translate text embeddings into realistic images through a generator-discriminator interplay, enhancing image diversity and quality. Contrarily, GANs face challenges such as potential mode collapse, the intricacy of generating fine-grained details, and a dependence on high-quality text embeddings. These factors can collectively influence the realism and diversity of the generated images. Our approach involves leveraging GANs to generate realistic images from text, thereby facilitating the creation of diverse and high-quality image datasets. This, in turn, contributes to enhancing the performance of downstream tasks through improved training.

Table 3.1: Literature Review Summary Table for Data Augmentation

This table explains the features, relevance and limitations of different techniques for Data Augmentation

Application	Features	Relevance to Application	Limitations
Text-to-Text Augmentation [2]	Word Deletion, Synonym replacement, Embedding swap, Inner swap random and BERT-Masked LM swap	Enriches the caption dataset by generating diverse variations and improving model generalization	Potential loss of context, over-reliance on language models.
Stable Diffusion Model [3]	Preserves the semantic meaning and diversity of the images.	Controlled and coherent generation of diverse images corresponding to text embeddings	Challenges in capturing complex image features.
GAN [4]	Generator-Discriminator Interplay	Aiding in the creation of diverse and high-quality image datasets for training and improving performance in downstream tasks.	Potential mode collapse and difficulty in generating fine-grained details.

3.2.2 Image Captioning

Over the past few years, image captioning has become a pivotal intersection between computer vision and natural language processing. Its objective is to produce descriptive and contextually relevant captions for images, empowering machines to comprehend and articulate visual content. Traditional approaches in image captioning relied on recurrent neural networks, particularly LSTM and Bi-LSTM models.

Several algorithms for image captioning are discussed:

3.2.2.1 CNN - LSTM based Image captioning model (CL):

Accoridng to the author Mingxing Tan and Quoc V. Le. [7]. In our LSTM-based image captioning model (CL), we use a pretrained EfficientNet for image feature extraction. These features go through LSTM layers to generate the output text sequence. The input image becomes a 2048-dimensional feature vector as shown in Figure 3.3, then passes through a linear layer to match word embedding dimension (512). This feature vector serves as input for the LSTM decoder, starting decoding. It sequentially generate captions based on visual features but limited context capture, struggles with long-range. We use a greedy decoder for token prediction at each step.

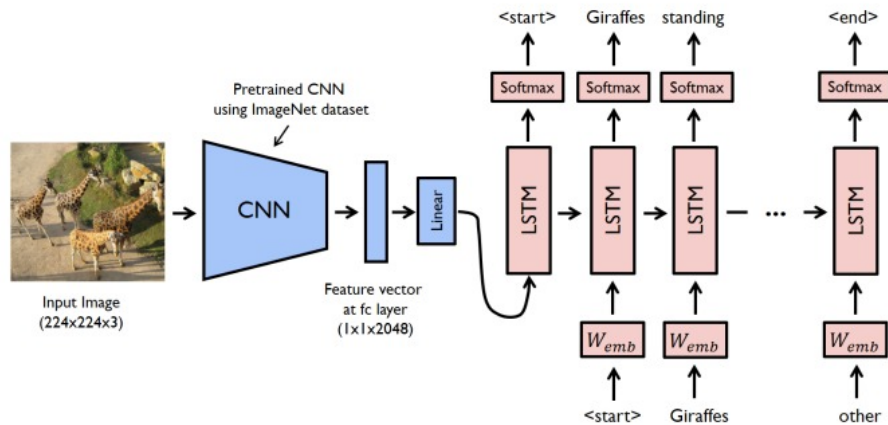


Figure 3.3: Architecture of our CNN-LSTM model

CNN-LSTM model that extract features from images using a pre-trained CNN and generate captions using an LSTM model.

3.2.2.2 CNN - Bi-LSTM based Image captioning model (CBL):

According to the author Mingxing Tan and Quoc V. Le. [7], Bi-LSTM closely resembles the CNN-LSTM model, with the key distinction being the use of a Bi-LSTM layer instead of an LSTM layer in the decoder. Generally, research has indicated that employing a Bi-LSTM layer in an encoder, rather than a LSTM layer, is more advantageous.

By utilizing bidirectional Long Short-Term Memory (LSTM) units, this system captures contextual information from both preceding and subsequent words in the caption. This capability enhances the contextual understanding and quality of the generated image descriptions. However, it faces challenges in modeling global context and coping with lengthy sequences, which could potentially result in an incomplete grasp of image-context relationships. Our approach will use its ability to capture long-range dependencies and context in captions bidirectionally, leading to more accurate and contextually informed descriptions for the corresponding images.

3.2.2.3 Transformer Models

According to the author Deepika Kumar [8], Transformer models as shown in Figure 3.4 are now the leading approach in image captioning. Their proficiency in capturing global context and long-range dependencies makes them ideal for complex visual scenes and coherent captions.

They have outperformed LSTM and RNN methods and represent the forefront of image captioning. These models excel at capturing long-range dependencies but require substantial data for training. Our approach will use new state-of-the-art models, to provide contextually relevant and coherent image captions.

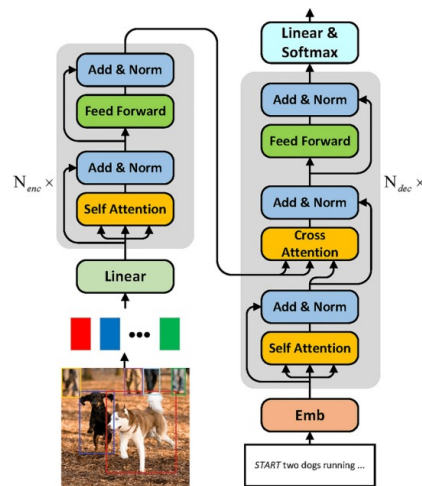


Figure 3.4: Architecture of a transformer based image captioning model

Transformer models encode image features and decode them into a caption one word at a time using self-attention and cross-attention.

Evaluation Metric: According to the author Kishore Papineni, [9] BLEU, or Bilingual Evaluation Understudy, assesses machine-generated text quality by comparing it to human-translated text. Note that BLEU doesn't consider grammar. In image retrieval, captions from an image captioning model guide the search. In text-based retrieval, users input keywords; in content-based, images are used for searching. We will use the BLEU to measure the accuracy of our algorithms.

$$\text{BLEU} = \min \left(1, \frac{\text{output length}}{\text{reference length}} \right) \left(\prod_{i=1}^4 \text{precision}_i \right)^{\frac{1}{4}} \quad (3.2)$$

Table 3.2: Sample Literature Review Summary Table for Image Captioning

This table explains the features, relevance and limitations of different techniques for Image Captioning

Application	Features	Relevance to Application	Limitations
CNN-LSTM [7]	Sequentially generate captions based on visual features. Suitable for encoding sequential data.	Fundamentally applied in early image captioning	Limited context capture, struggles with long-range dependencies.
Bi-LSTM [7]	Enables capturing contextual information from both past and future words in the caption	its Ability to capture long-range dependencies and context in captions bidirectionally	Modeling global context and difficulty in handling very long sequences
Transformer Models [8]	Excel at capturing complex relationships between images and text. Efficiently handle long-range dependencies	New state-of-the-art models, providing contextually relevant and coherent image captions.	Require substantial data for training, computationally intensive.

3.2.3 Content-Based Image Retrieval

Efficiently accessing specific images from extensive and diverse databases is imperative. Content-Based Image Retrieval (CBIR) operates by retrieving images based on visual features, eliminating the need for metadata or keywords. In this process, each image in the database has its features extracted and is then compared to the features of the query image. Typically, systems perform feature extraction [10] as a preprocessing step, capturing global image features such as corners, edges, ridges, colors, textures, intensity, and more.

Several algorithms are discussed in matching the features of images:

3.2.3.1 Convolutional Autoencoder

Eman S. Sabry [11], presents a study on image retrieval using convolutional autoencoder, InfoGAN, and Vision Transformer unsupervised models. An autoencoder in machine learning compresses data by transforming input into similar output. It uses neural networks to learn compression and decompression functions automatically. The proposed architecture, illustrated in Figure 3.5, combines convolutional neural networks (CNNs) to enhance image retrieval. The encoder condenses images into feature descriptors, while the decoder reconstructs them. These descriptors are then compared using a nearest-neighbor algorithm to find similar images. This method efficiently represents image features in low-dimensional

vectors, improving retrieval accuracy across different image categories.

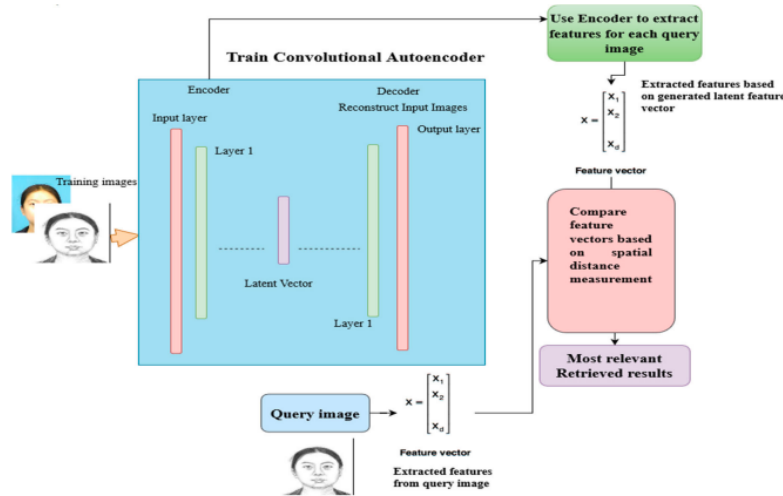


Figure 3.5: Image retrieval using convolutional autoencoder.

This figure illustrates the sequential process undertaken by the model, explaining each step comprehensively.

Convolutional Autoencoders (CAEs) are good at finding important stuff in pictures. They're like detectives that look at every detail and figure out what's important. They can then store this important stuff in a smaller space, which is like keeping a tidy summary of the picture. However, sometimes they might miss out on some small details, and understanding everything they find can be a bit tricky. But, when it comes to finding similar pictures, they're super useful. They can quickly compare the important stuff they found in one picture with others, helping you find pictures that are alike. This makes them handy for things like searching for similar images on the internet or in big collections of pictures.

3.2.3.2 SIFT Algorithm

The SIFT algorithm, introduced by S. Jain and K. Pulaparthi [12] for keypoint detection and description as illustrated in Figure 3.6, has proven to be instrumental in image processing. Its key feature lies in the extraction of keypoints, offering independence from scale variance and rotation variance.

Comprising primarily five steps to enhance its results: Scale Space Extrema detection, Keypoint Localization, Orientation Assignment, Feature description, and Feature matching, the SIFT algorithm is a powerful tool. However, its computational complexity and sensitivity to image transformations impose limitations on its efficiency and accuracy, especially for real-time Content-Based Image Retrieval (CBIR) on large-scale datasets. Nevertheless, our approach can leverage its robust and distinctive local image features, enabling accurate matching and retrieval of similar images based on their keypoints and descriptors.

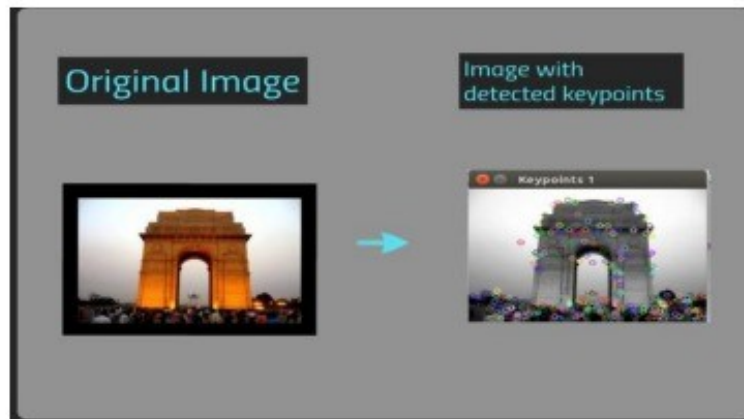


Figure 3.6: Detected Keypoints using SIFT

This figure represents the keypoints of an image detected by SIFT algorithm

3.2.3.3 SURF Algorithm

As per K. Velmurugan [13], the SURF algorithm serves as a scale and rotation-invariant interest point detector and descriptor, known for its exceptional computational speed. Utilizing Integral images to enhance processing speed, key points are identified through a Fast-Hessian matrix. The descriptor, capturing the distribution of Haar-wavelet responses within the interest point neighborhood, sees improved performance with the use of an intermediate image representation called the Integral Image. Rapidly computed from the input image, the integral image accelerates the calculation of any upright rectangular area.

The SURF descriptor extraction from an image occurs in two steps. First, an orientation is assigned based on the information within a circular region surrounding the detected interest points. In the second step, the region is systematically divided into smaller square sub-regions, and a few simple features at regularly spaced sample points are computed for each sub-region.

It boasts several times the speed of SIFT and is purportedly resilient against various image transformations compared to SIFT but it may struggle with handling scale variations and complex image deformations. Our approach can use it due to its speed and robustness in detecting and describing key features, enabling efficient and accurate matching for image similarity and retrieval.

3.2.3.4 Interactive Genetic Algorithm

In a work by Chin-Chin Lai [14], an interactive genetic algorithm (IGA) was introduced to narrow the gap between retrieval results and user expectations. This approach incorporates color attributes such as mean value, standard deviation and image bitmap. Additionally texture features like entropy based on

the gray level co occurrence matrix and the edge histogram are utilized. Comparative analysis with other approaches revealed superior results with this method.

On the other hand, the effectiveness of IGA in CBIR can be limited by the subjectivity of user feedback, potential biases and the challenge of accurately capturing complex user preferences that leads to suboptimal retrieval outcomes. We can use them by incorporating user preferences to guide the retrieval process that results in personalized and more accurate image searches aligned with individual user tastes and requirements.

Evaluation metric: According to E. E. Eldarova, [15] SSIM which stands for Structural Similarity Index Measure is a metric used to quantify the similarity between two images. It is widely used in the field of image processing and computer vision to evaluate the quality of images especially in tasks like image compression, image restoration and image enhancement. The SSIM index is calculated based on three components: luminance, contrast, and structure. Let's break down these components:

- **Luminance:** This component measures the similarity in the brightness of corresponding pixels between the two images. It is computed using the mean intensity of the pixels.
- **Contrast:** Contrast refers to the difference in intensity between pixels in an image. SSIM measures the similarity in contrast between corresponding pixels in the two images.
- **Structure:** Structure represents the correlation between neighboring pixels. SSIM evaluates how well the structures in the two images match.

The SSIM index is calculated by combining these three components into a single value that ranges from -1 to 1 where 1 indicates perfect similarity between the images.

The formula for SSIM is as follows:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C1)(2\sigma_{xy} + C2)}{(\mu_x^2 + \mu_y^2 + C1)(\sigma_x^2 + \sigma_y^2 + C2)} \quad (3.3)$$

Where:

- μ_x and μ_y are the means of the images x and y .
- σ_x^2 and σ_y^2 are the variances of the images x and y .
- σ_{xy} is the covariance of the images x and y .
- C_1 and C_2 are constants to stabilize the division with weak denominator.

Table 3.3: Sample Literature Review Summary Table for CBIR

This table explains features, relevance and limitations of different techniques for CBIR

Application	Features	Relevance to Application	Limitations
Autoencoder [11]	Uses convolutional layers for spatial pattern extraction, learns a compressed latent space representation, reconstructs images from latent space.	Extracts meaningful image features, enables efficient similarity comparisons, facilitates semantic understanding.	Loss of fine details, limited generalization, computational complexity, dependency on training data quality.
SIFT [12]	Scale Space Extrema, Keypoint Localization, Orientation Assignment	Robust and distinctive local image features based on their key points and descriptors.	Computational complexity and sensitivity to image transformations
SURF [13]	Forms a grid around the keypoint and subdivides each grid cell into smaller subgrids	Fast and robust in detecting and describing key features	Struggles with handling scale variations and complex image deformations
IGA [14]	User Interaction, Evolutionary Process and Real-Time Responsiveness	Incorporates user preferences to guide the retrieval process	The subjectivity of user feedback and the challenge of accurately capturing complex user preferences
K-Means [14]	Groups the training data into clusters and produces the clustering result as a learning codebook.	Enables clustering of image features, aiding in grouping similar images	Struggles with capturing complex image structures and may result in suboptimal clustering

3.3 Conclusion

In conclusion augmentation techniques are pivotal for enhancing machine learning models by diversifying both textual and image data. Text augmentation methods such as synonym replacement, word deletion, and embedding swap, implemented through frameworks like NLPAug significantly enrich the textual dataset. Similarly, image augmentation utilizes generative models like GANs and Stable Diffusion that proves powerful and generating novel images from given prompts and thereby enhancing datasets and model performance in various computer vision tasks.

The field of image captioning has evolved dramatically transitioning from conventional techniques like LSTM and Bi-LSTM to cutting-edge Transformer models. These advancements have revolutionized caption generation for images.

Moreover, K-Means is applied for vector quantization, showcasing its relevance. Additionally, methods like Convolutional Autoencoder, SIFT and SURF address challenges in image retrieval, emphasizing continuous progress and innovation within computer vision and its wide-ranging applications.

Chapter 4 Software Requirement Specifications

The following section specifies all the requirements with their uses case description. A data diagram with data dictionary is also included in this section.

4.1 List of Features

A list of key features for the users of the system are listed below:

4.1.1 Image captioning

Users can input an image and receive descriptive and relevant captions that accurately depict the contents and context of the image.

4.1.2 Image Retrieval

Users can upload an image to initiate a search and retrieve visually similar or related images from the system's database.

4.1.3 Historical Record-keeping

Users can view their past queries and their results.

4.2 Functional Requirements

The functional requirements of the application are listed below.

- Users can sign up for creating a new account.
- Users can sign in to the system.
- Users can sign out of the system.

Users can view the buttons of image captioning and image retrieval.

Image captioning:

- User can upload one image, multiple images(atmost 50 images) or a folder as input at a time.
- The application will only input .jpg and .png files.
- The application will preprocess the input images to enhance clarity and reduce noise if necessary.
- The application will display the inputs and the captions alongside.

- The application will output at least two captions for each image.
- For multiple images or a folder, the users can choose to store the captions in a text file that can be downloaded.
- The application will save the user's past records and their captions.
- Users can delete some past records from its history.

Image retrieval:

- User can upload one image at a time.
- The application will only input .jpg and .png file.
- The application will preprocess the input image to enhance clarity and reduce noise if necessary.
- The application will output at least two similar images.
- Users can download one output image or all output images at a time.
- The application will save the user's past records and their results.
- Users can delete some past records from its history.

4.3 Quality Attributes

The quality of the system will be analyzed based on the completion of all the listed functional requirements. In addition, we will consider factors contributing to reliability, usability, scalability, security, and maintainability. The aim is to make the system interface simple and effective. Therefore, the learning ability of the user would be taken into consideration so that it doesn't take longer for one to become used to the system's interface.

4.4 Non-Functional Requirements

The system will meet the following non-functional requirements to optimize its performance and deliver the best results.

4.4.1 Usability

Our system's interface is designed for easy learning and navigation. Buttons, headings, and help/error messages are straightforward and easily comprehensible. Users can grasp the interface without the need for special training; even beginners can start using it after just a few interactions.

4.4.2 Performance

The application will prioritize efficiency and optimal performance by leveraging Django framework, renowned for its runtime server capabilities and performance optimization. Our focus will be on ensuring exceptional performance with minimal turnaround time.

4.4.3 Security

The main concern will be security. Every user accessing our system will have authorized access, and robust measures will be in place to defend against potential threats like SQL injection. Protection of private data is prioritized, ensuring that only information permitted by the specific user is exposed.

4.4.4 Reliability

The application will consistently function as expected across different operating systems, hardware configurations, and versions.

4.4.5 Extensibility

A modular approach will be employed, ensuring that replacing or adding new components can be seamlessly accomplished without complications at any given time.

4.5 Assumptions

Following are the assumptions considered to enable the use of the system:

- User should have a device capable of opening application.
- Users should have a stable browser (Chrome, Firefox, Edge, Opera).
- User should be smart enough to interact properly with the interface of the application.

4.6 Use Cases

The use case descriptions for all features of system are listed below.

4.6.1 Sign Up

Table 4.1: Use case Description for Sign Up

Outlined in this use case description are the steps and interactions involved in the user sign-up process.

Name	Sign Up		
Actors	Regular Users		
Summary	On the sign-up form, the user is required to provide a username and set a password.		
Pre-Conditions	The user must not already exist in the database records.		
Post-Conditions	Upon successful establishment of the user’s session, they will be redirected to the sign-in page.		
Special Requirements	User name and password must be less than 50 characters long.		
Basic Flow			
Actor Action		System Response	
1	The user navigates to the sign-up page.	2	The displayed sign-up page prompts the user to enter their username and password.
2	The user inputs a valid username and password.	4	The system stores the entered username and password in the database and subsequently redirects the user to the sign-in page.
Alternative Flow			
3	The user inputs an invalid username or password.	4-A	The system generates an error message stating, ”Usernames and passwords must have a length of fewer than 50 characters.”

4.6.2 Sign In

Table 4.2: Use case Description for Sign In

Outlined in this use case description are the steps and interactions involved in the user sign-in process.

Name	Sign in		
Actors	Regular Users		
Summary	The user will Sign in with username and password to use the application.		
Pre-Conditions	The Sign in page should be displayed on screen.		
Post-Conditions	User will successfully login and redirected to the home page		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user navigates to the sign-in page.	2	The displayed sign-in page prompts the user to enter their email and password.
2	The user inputs a valid email and password.	4	Upon confirming the username and password, the system creates a user session and guides the user to the home page.
Alternative Flow			
3	The user inputs an invalid username or password.	4-A	The system issues an error message stating, "Incorrect username or password entered."

4.6.3 Sign Out

Table 4.3: Use case Description for Sign out

Outlined in this use case description are the steps and interactions involved in the user sign-out process.

Name	Sign Out		
Actors	Regular Users		
Summary	The user intends to log out.		
Pre-Conditions	The user must be currently logged in.		
Post-Conditions	The user will successfully log out.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user clicks on the sign out button.	1-A	The system will display a confirmation message asking, "Are you sure you want to log out?" with options for both Yes and No.
1-A	User will select "Yes".	2	System will redirect the user to sign in page.
Alternative Flow			
1-A	User selects "No" option.	3	System will cancel logout request.

4.6.4 Home Page

Table 4.4: Use case Description for Home Page

This use case description details the interactions and functionalities accessible to users on the home page.

Name	Home Page		
Actors	Regular Users		
Summary	The user wants to generate captions for an image/images or wants to retrieve more images for a given Image .		
Pre-Conditions	The user must be currently logged in.		
Post-Conditions	The user will obtain the necessary captions for the image/images or the search related images.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	After the user logs in.	2	The system will navigate to the home page.
2	User will select "Captioning".	4	System will redirect the user to image Captioning page.
Alternative Flow			
3	User will select "Retrieval" option.	5	System will show the image retrieval page.

4.6.5 Image Uploading for Image Captioning

Table 4.5: Use case Description for Image Uploading for Image Captioning

In this use case description, the steps and interactions for uploading an image for captioning are outlined.

Name		Image Uploading for Image Captioning	
Actors		Regular Users	
Summary		The users can select images from storage.	
Pre-Conditions		The user needs to be on image captioning page	
Post-Conditions		The captions will be displayed alongside the images.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user opens the image captioning page.	2	The page is displayed where user can upload images.
2	Click on “Import from storage” button	3	System will open the directory.
4	User will select the image from directory.	2	Selected images will be passed to image captioning module and captions will be generated.
Alternative Flow			
2-A	User does not select any photo.	2-B	System prompt the message “Please select photo”

4.6.6 Download Captions File

Table 4.6: Use case Description for Downloading Captions File

In this use case description, the steps and interactions for downloading output file are outlined.

Name		Download Captions File	
Actors		Regular Users	
Summary		The users can download the text file of captions.	
Pre-Conditions		The user must be given multiple images as input.	
Post-Conditions		The file will be successfully downloaded.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user selected multiple images for input.	2	The system wrote captions of all images in a file.
2	The user will click the download button to download the text file.	4	System will download the file.
Alternative Flow			
3	User uploaded only one image as input	4-A	System will not generate any output file.

4.6.7 Image uploading for Image Retrieval

Table 4.7: Use case Description for Image Uploading for Image Retrieval

In this use case description, the steps and interactions for uploading an image for retrieval are outlined.

Name		Image Uploading for Image Retrieval	
Actors		Regular Users	
Summary		The users can select image from storage.	
Pre-Conditions		The user must be on image retrieval page	
Post-Conditions		The related images will be displayed alongside the image.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user opens the image retrieval page.	2	The page is displayed where user can upload images.
2	Click on “Import from storage” button	3	System will open the directory.
4	User will select the image from directory.	2	Selected image will be passed to image retrieval module and related images will be generated.
Alternative Flow			
4-A	User does not select any photo.	2-A	System prompt the message “Please select photo”

4.6.8 Download Output Images

Table 4.8: Use case Description for Downloading Output Images

In this use case description, the steps and interactions for downloading output images are outlined.

Name		Download Output Images	
Actors		Regular Users	
Summary		The users can download the output images in case of retrieval.	
Pre-Conditions		The user must uploaded an image as input.	
Post-Conditions		The images will successfully be displayed.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user uploaded one image.	2	The system displayed similar images.
2	The user will click the download button to download the related images.	4	System will download the output images.

4.6.9 History Log

Table 4.9: Use case Description for History Log

This use case description outlines the steps and interactions involved in viewing history

Name	History Log		
Actors	Regular Users		
Summary	The users can view their past search images along with the results.		
Pre-Conditions	The user must have used our system before.		
Post-Conditions	The past actions will successfully be displayed.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user will click on the history button.	2	The system will display his past searches of one month alongwith date and time
2	The user will click the download button to download some specific past results.	4	System will download the results.
Alternative Flow			
2-A	User does not have any past records.	4-B	System prompt the message “No past Records!”

4.6.10 History Deletion

Table 4.10: Use case Description for History Deletion

This use case description outlines the steps and interactions involved in deletion of history

Name		History Deletion	
Actors		Regular Users	
Summary		The users can delete their past search images along with the results.	
Pre-Conditions		The user must have history records.	
Post-Conditions		The selected past actions will successfully be deleted.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user will select some records.	2	The system will display the selected records and ask ”Are you sure to delete the selected records”
2-A	The user will click ”Yes”.	3	System will delete the results.
Alternative Flow			
2-A	The user will select ”No”.	4	System will go back to history page.
2-C	The user will select ”Edit”.	1	System will go back to history page with already selected images.

4.7 Hardware and Software Requirements

Here are the software and hardware prerequisites for the proper functioning of our application:

4.7.1 Hardware Requirements

For the application to run smoothly the following requirements are necessary else there are chances that the application can lag.

- Laptop/Desktop with atleast 10GB VRAM and a GPU infrastructure and 64-bit operating system.
- Minimum 1GB disk space is required.

4.7.2 Software Requirements

For the application to be developed, the following softwares is needed:

- **Visio:** A tool for system modeling
- **Python:** For AI integration with web apps.
- **Chrome:** Browser for web apps
- **Google Colab:** Editor for users to access GPU-accelerated resources for computationally intensive tasks and have a cloud-based architecture.
- **Django:** Python web framework
- **TailwindCSS:** CSS framework packed with classes.
- **Figma:** Designing and prototyping of the web app.
- **VScode:** Editor for the development due to extensive available extensions.
- **XAMPP:** Editor to create a database.

4.8 Graphical User Interface

The below-specified GUI visually represents some core features of the ImageSense website.

4.8.1 Sign up page

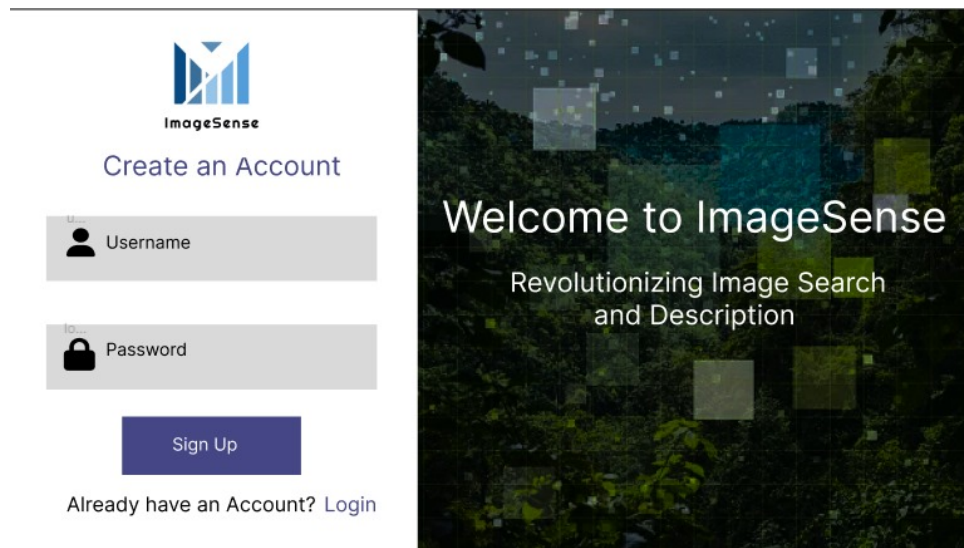


Figure 4.1: Sign up page

All Users can Sign up to application using this page

4.8.2 Sign in page

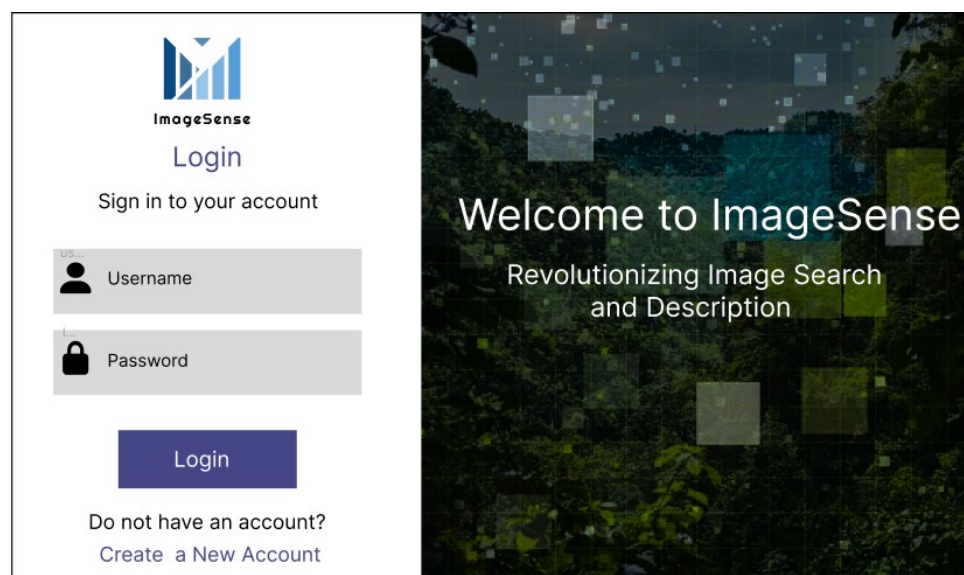


Figure 4.2: Sign in page

All Users can Sign in to application using this page

4.8.3 Home Screen

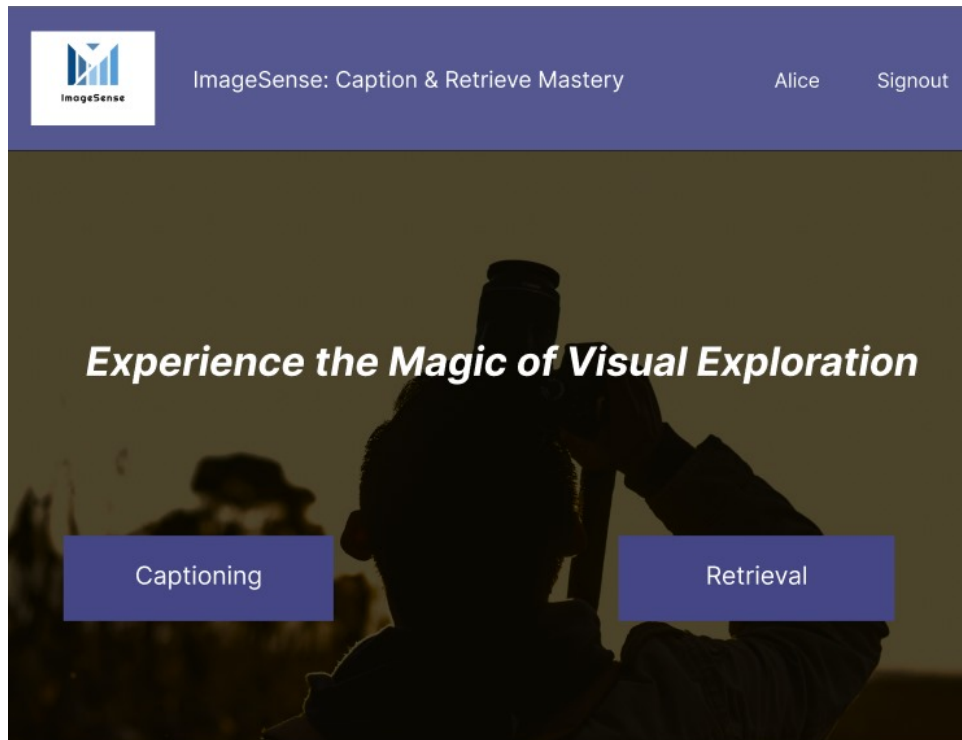


Figure 4.3: Home Screen

This is homepage of regular user

4.8.4 Upload Images to Generate Captions

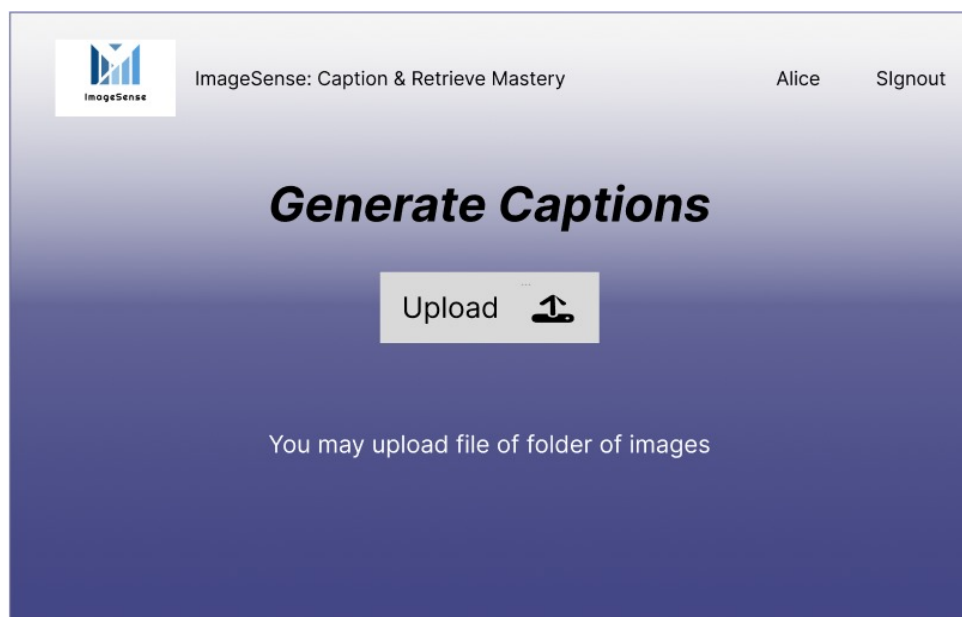


Figure 4.4: Upload Images to Generate Captions

Users can upload images to generate captions

4.8.5 Captions of Single Image



Figure 4.5: Caption of Single Image

Users can view captions of an image

4.8.6 Captions of Multiple Images

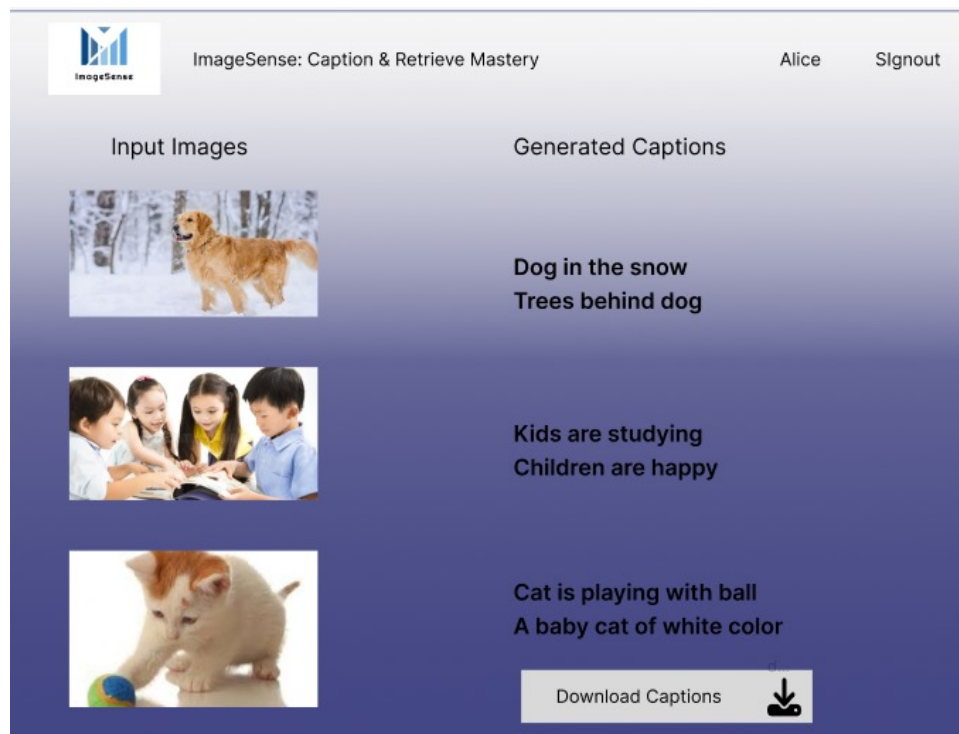


Figure 4.6: Captions of Multiple Images

Users can view and download captions of multiple images

4.8.7 Upload Image for Image Retrieval

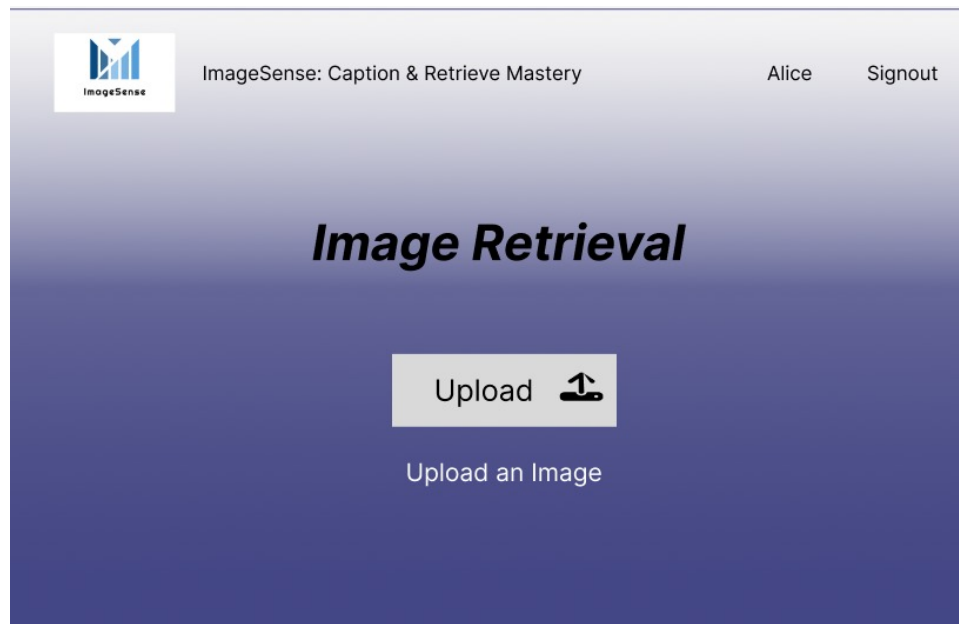


Figure 4.7: Upload Image for Image Retrieval

Users can upload image to generate related images

4.8.8 Output Similar Generated Images

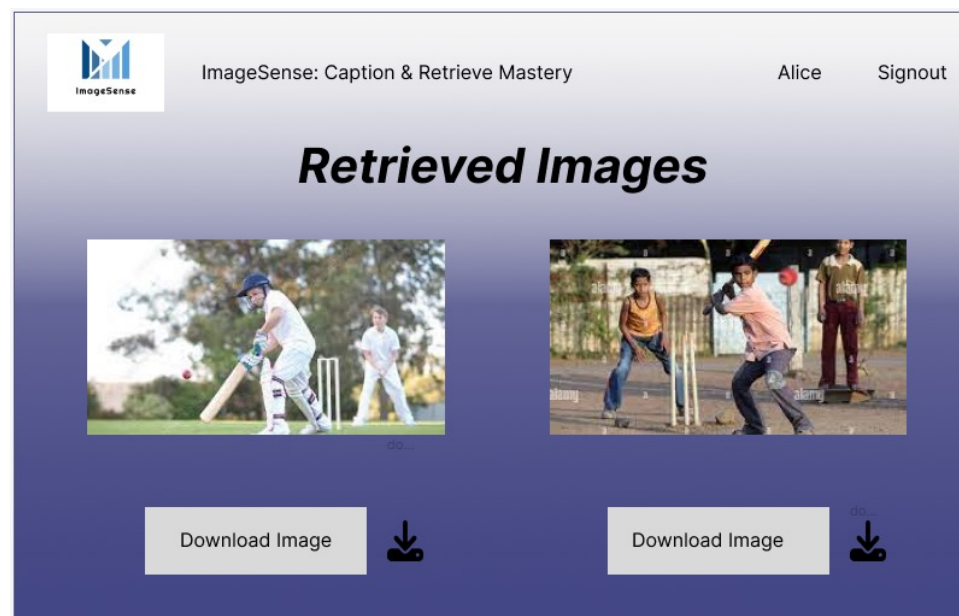


Figure 4.8: Output Similar Generated Images

Users can view related images

4.8.9 History Log



Figure 4.9: History Log

Users can view all past records

4.8.10 History Log of Image Captioning

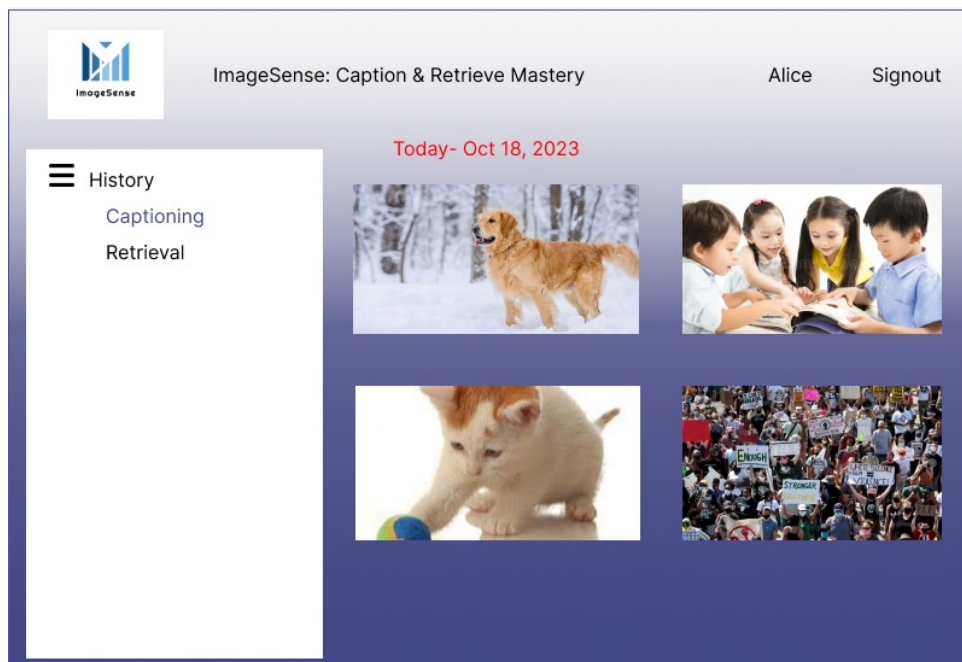


Figure 4.10: History Log of Image Captioning

Users can view all past records related and their results to image captioning

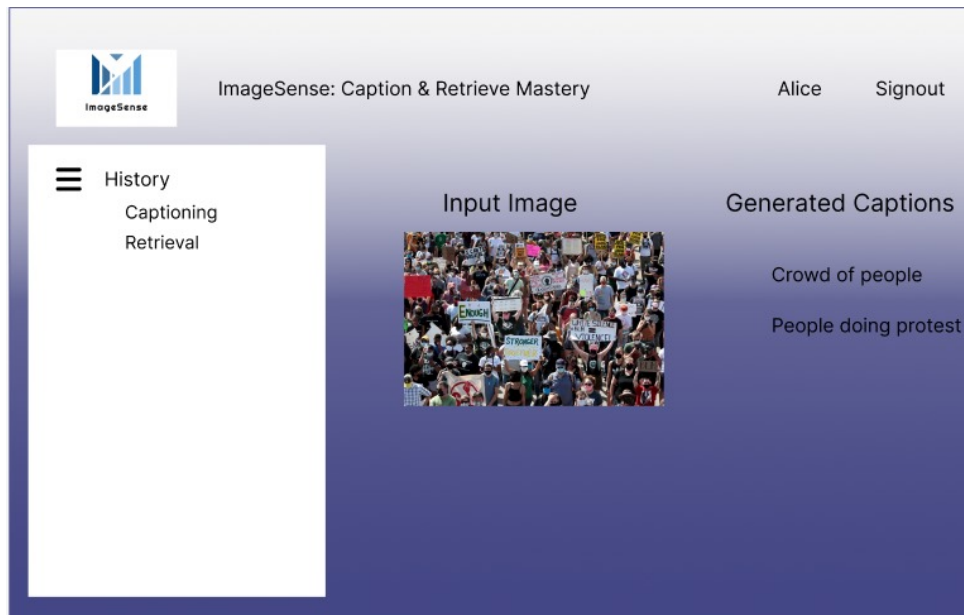


Figure 4.11: History of Image Captioning

Users can view captions of past images by clicking on the image

4.8.11 History Log of Image Retrieval

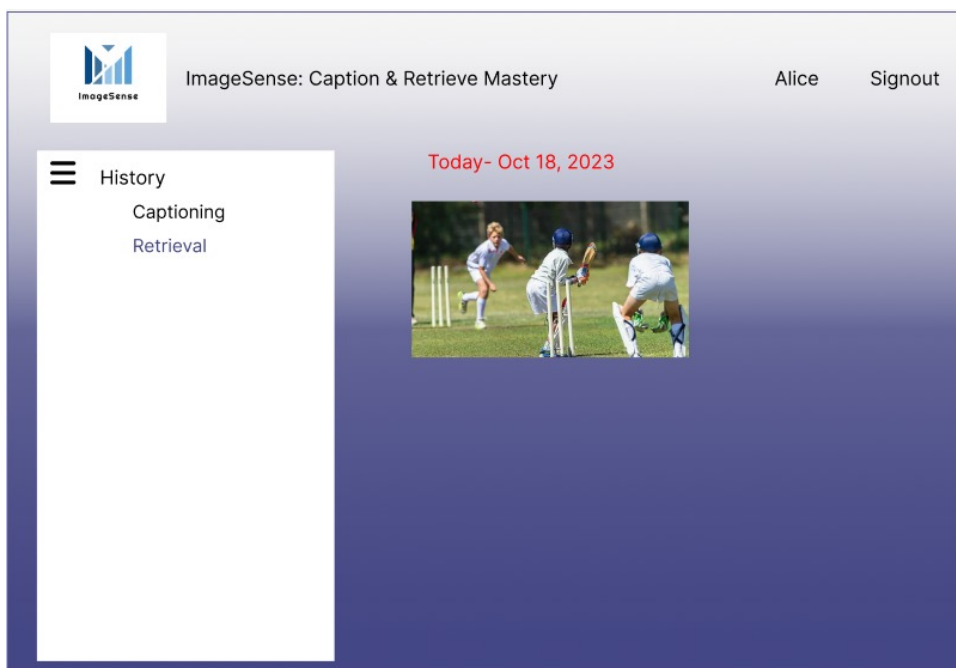


Figure 4.12: History Log of Image Retrieval

Users can view all past records and their results related to image retrieval

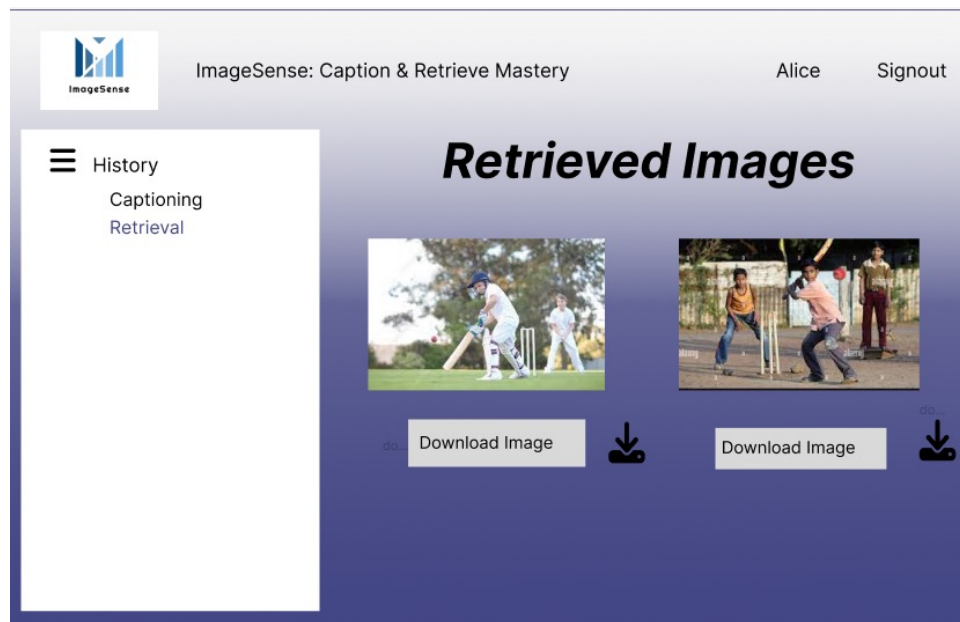


Figure 4.13: History of Image Retrieval

Users can view related images of past images by clicking on the image

4.8.12 History Deletion

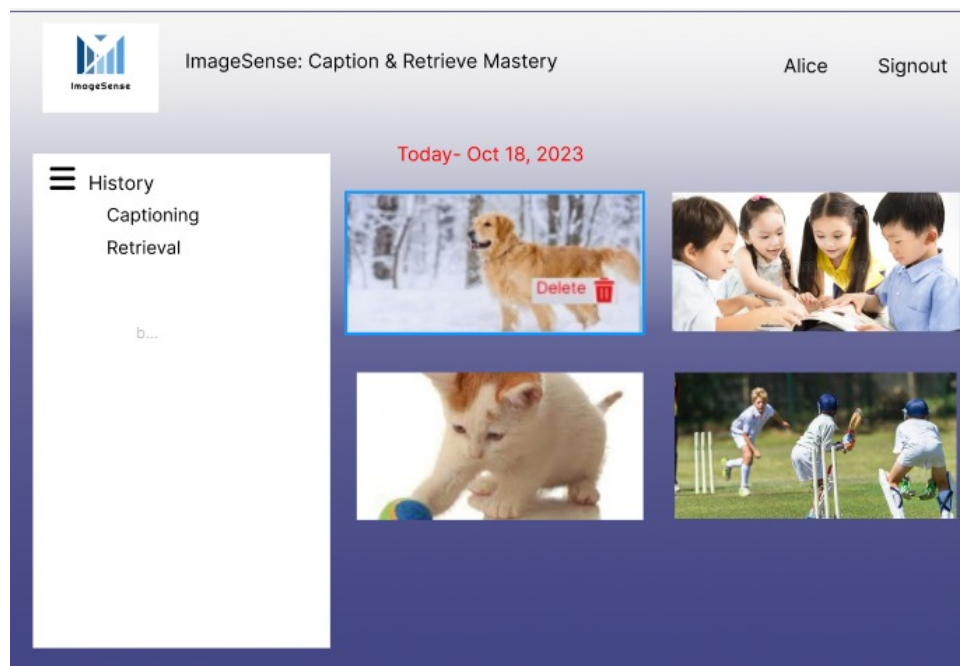


Figure 4.14: History Deletion

Users can delete any past record by just right clicking on it.

4.9 Database Design

The following section displays the entity relationship diagram with data dictionary.

4.9.1 ER Diagram

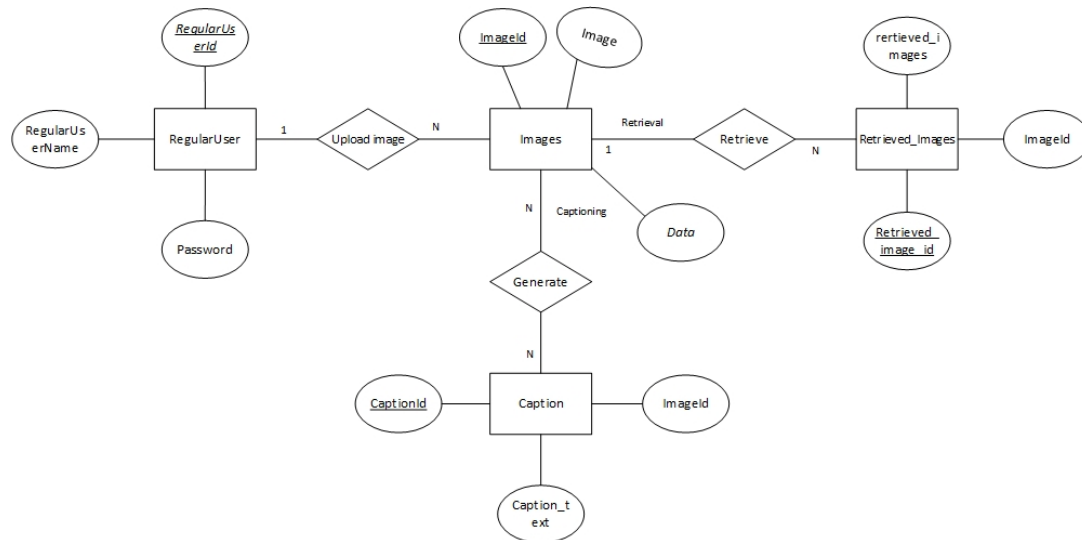


Figure 4.15: ER Diagram

The figure above shows entities and the relationship between them

4.9.2 Data Dictionary

Following is the Data Dictionary for ImageSense:

Table 4.11: RegularUser

A table describing attributes and their descriptions for a regular user

Column Name	Data Type	Column Definition	Nullable	Unique	Column is PK/FK
RegularUserId	int	Identifier for Regular user	No	Yes	PK
RegularUserName	varchar(50)	Full name of Regular User	No	No	-
Password	varchar(50)	Authentication of Regular User	No	No	-

Table 4.12: Images

A table describing attributes and their descriptions for an image

Column Name	Data Type	Column Definition	Nullable	Unique	Column is PK/FK
ImageId	int	Authentication of Image	No	Yes	PK
Image	BLOB	Images entered by Regular Users	No	No	-
Date	DATE	Date on which images are entered by Regular Users	No	No	-

Table 4.13: Caption

A table describing attributes and their descriptions for a caption

Column Name	Data Type	Column Definition	Nullable	Unique	Column is PK/FK
CaptionId	int	Authentication of Caption	No	Yes	PK
ImageId	int	Authentication of Image	No	No	FK
CaptionText	varchar(50)	Caption of Image	No	No	-

Table 4.14: RetrievedImages

A table describing attributes and their descriptions for retrieved images

Column Name	Data Type	Column Definition	Nullable	Unique	Column is PK/FK
RetrievedImageId	int	Authentication of Image	No	Yes	PK
ImageId	int	Authentication of Image	No	No	FK
RetrievedImages	BLOB	Relevant Images	No	No	-

4.10 Risk Analysis

The following section analyzes the risk involved in the system regarding different domains.

4.10.1 Technical Risks

The project's requirements are nearly settled, and significant alterations aren't anticipated. This means the system is not at high risk of experiencing sudden requirement shifts. To streamline the system's structure, each module will be handled independently and subjected to comprehensive testing. regarding different domains.

4.10.2 Programmatic Risks

The rapidly evolving field of AI models, particularly deep learning ones, are susceptible to adversarial attacks, wherein malicious individuals deliberately alter inputs to deceive the model's predictions.

4.11 Conclusion

In conclusion, this chapter describes a comprehensive list of features, functional requirements and quality attributes essential for the successful development of the software. While outlining non-functional requirements, assumptions and use cases, the document provides a thorough insight into the project's scope and constraints. Additionally, the chapter highlights critical considerations such as hardware and software requirements, graphical user interface, database design and risk analysis that lays the foundation for a well informed and structured software development process.

Chapter 5 High-Level and Low-Level Design

This following section specifies the high-level and low-level designs of the project. It includes the system overview, the design considerations, system architecture, architectural strategies, class diagram, sequence diagrams and the policies and tactics related to the project.

5.1 System Overview

In this section, we will discuss the overall overview of the system. This will include all steps starting from designing the system architecture, augmenting dataset, training the model, feeding the data to the system and the output presented to the user in response.

5.1.1 System Design

Following describes the compositional structure of the system.

5.1.1.1 Image Uploading

The image uploading module functions as the initial point for users to submit images. As users upload images, the system utilizes effective data handling processes to guarantee a smooth transfer of visual data. It also incorporates error-handling measures and feedback mechanisms to assist users during the upload. The system architecture is structured to manage concurrent uploads ensuring a responsive experience.

5.1.1.2 Feature Extraction

Then the features of images will be extracted for deriving meaningful representations from images. These features play a key role in image captioning by aiding in understanding image content and context for generating descriptive captions. Additionally in image retrieval, these features are essential for quantifying visual similarities among images that forms the foundation for efficient retrieval algorithms.

5.1.1.3 Results Formulation

This stage involves compiling generated captions and retrieved images. In image captioning, the formulated results encompass descriptive captions by offering users contextually rich interpretations of submitted images. Regarding image retrieval, the results comprise visually similar images based on user input. This section emphasizes the clarity and relevance of the information delivered to users as the final output of the system.

5.1.1.4 Results Storage

This stage involves storing generated captions and similar images in database for future reference and retrieval.

5.1.2 Data Collection

We will utilize the Flickr8k dataset and implement two types of data augmentation to enhance diversity: text-to-text augmentation and text-to-image augmentation. For text-to-text augmentation, techniques such as Word Deletion, Synonym Replacement, Embedding Swap, Inner Swap (random), and BERT-Masked LM Swap will be applied. On the other hand, text-to-image augmentation involves leveraging the Stable Diffusion Model to introduce diversity into our dataset because even though the best models require large and sufficient amount of training data to perform well.

5.1.3 Model Training

In our approach to image captioning, we opt for training a Transformer-based model on the augmented dataset. This choice is driven by the model's ability to capture long-range dependencies efficiently through parallel image encoding, facilitated by a self-attention mechanism. Conversely, for image retrieval, we employ the Scale-Invariant Feature Transform (SIFT) due to its robustness in handling changes in scale and rotation. This two-pronged strategy ensures the effectiveness of our system in generating descriptive captions and retrieving visually similar images.

5.2 Design Considerations

This section describes the assumptions and dependencies of the system, the general constraints, the goals and guidelines, and the development methods.

5.2.1 Assumptions and Dependencies

The project makes use of the following assumptions:

- Implemented models efficiently operate within specified computational and storage constraints.
- End-users have access to the internet and modern computing devices
- Adequate hardware resources, are available for ML model training and execution.
- High-quality training data is utilized for image captioning and CBIR tasks.

5.2.2 General Constraints

Our project, ImageSense: Caption and Retrieve Mastery, operates within certain global constraints that shape the design of our software. These constraints encompass various aspects:

5.2.2.1 Biased Data Constraint

Limited Biased high-quality training data poses a challenge for effective models in image captioning and content-based image retrieval.

5.2.2.2 Input Limit Constraint

There is an input limit in our project. It has capability to process only 50 images at a time.

5.2.2.3 Language Constraint

Our captions will only be available in English language. No any other language is supported by our system.

5.2.2.4 Performance Constraint

High computation power is required for efficient performance.

5.2.2.5 Accessibility Constraint

Application developed will not be publicly accessible.

5.2.3 Goals and Guidelines

The overarching goals and guidelines influencing the design of the system's software include:

5.2.3.1 Goals

- System will be user-friendly.
- System will focus on accuracy rather than resource consumption.
- System will boost effectiveness, precision, and quality of journalists' tasks.
- Performance of system in generating captions will be fast but image retrieval task will be time consuming because of high computation requirement.

5.2.3.2 Guidelines

- User must ensure that images uploaded for captioning or retrieval are of high quality for efficient results.
- User must respect the upload limits imposed by the system.

5.2.4 Development Methods

For our project development we will be using Scrum development methodology. Scrum is an agile development methodology that allows flexibility and continuous adaptation throughout the development cycle. Since, we have to develop this system based on the feedback from our advisor, it is obvious that many modifications might be required. Work will be divided into several sprints, each of which will have a deadline.

5.3 System Architecture

The system's core responsibilities are generating descriptive and contextually relevant captions that accurately portray the contents of an image, and retrieving visually similar or related images.

To achieve these multifaceted responsibilities, the system is structured around a Microservices architecture as shown in Figure 5.1. This architectural approach ensures modularity and flexibility, allowing the software to adapt to the dynamic demands of content creation. The architecture is organized into two key modules, as depicted in the high-level architecture diagram.

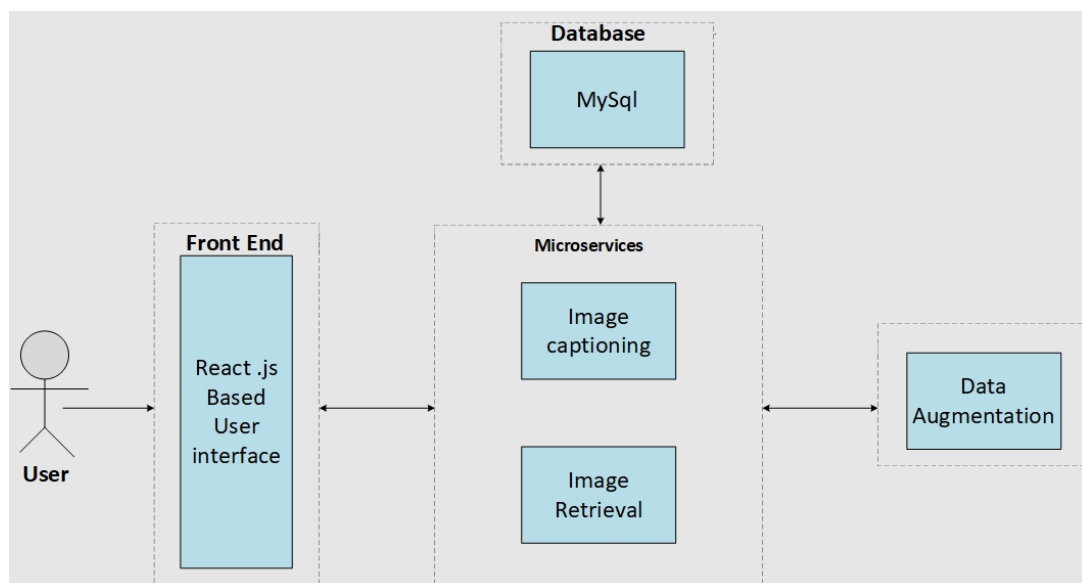


Figure 5.1: Architecture Diagram

The figure above shows the complete architecture of the system

5.3.1 Subsystem Architecture

The architecture is further divided into two distinctive sub-architectures, specifically tailored for image captioning and image retrieval as shown in Figure 5.2 and Figure 5.3 respectively. Both modules are using Pipe and Filter architecture. The detailed discussion of the modules is presented as follows:

5.3.1.1 Image Captioning

- **Preprocessing:** Preprocesses the images by resizing, normalizing, and converting them to a suitable format for the image encoder and preprocess the captions by tokenizing, cleaning, and converting them to a suitable format for the language decoder.
- **Encoder:** Captures visual features from the input image and expressing them as a series of vectors.
- **Attention Mechanism:** Integrates an attention mechanism between the encoder and decoder to allow decoder to directs attention to pertinent segments of the visual features during the caption generation process.
- **Decoder:** Generates a caption word by word, considering both the visual features from the encoder and the previously generated words.
- **Loss Function:** Measures the difference between the generated caption and the real caption.
- **Optimization:** Updates the model parameters based on the loss function.

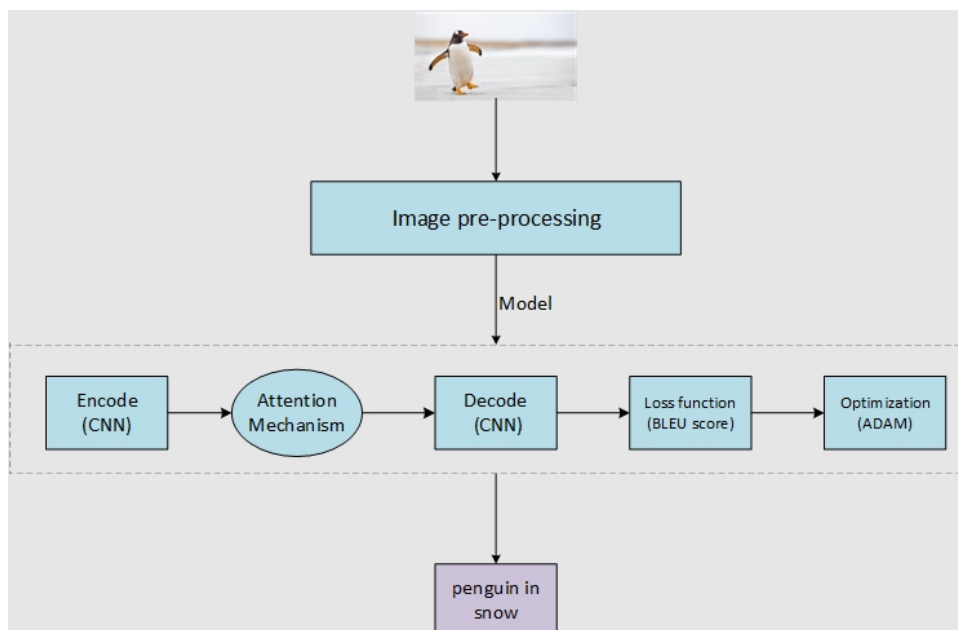


Figure 5.2: Image Captioning Subsystem Architecture Diagram

The figure above shows the subsystem architecture of the image captioning module of the system

5.3.1.2 Image Retrieval

- **Image Preprocessing:** Preprocesses the images by resizing, normalizing, noise reduction and converting them to a suitable format.
- **Feature Extraction:** Extracts scale-invariant, rotation-invariant, and illumination-invariant descriptors from keypoints in the image.
- **Feature Indexing:** Creates a visual vocabulary from SIFT descriptors and expresses each image as a histogram detailing the frequencies of visual words.
- **Similarity Calculation:** Calculates the similarity between the query image and each image in the database
- **Ranking and Retrieval:** Ranks the retrieved images based on their similarity scores and present the top results to the user.

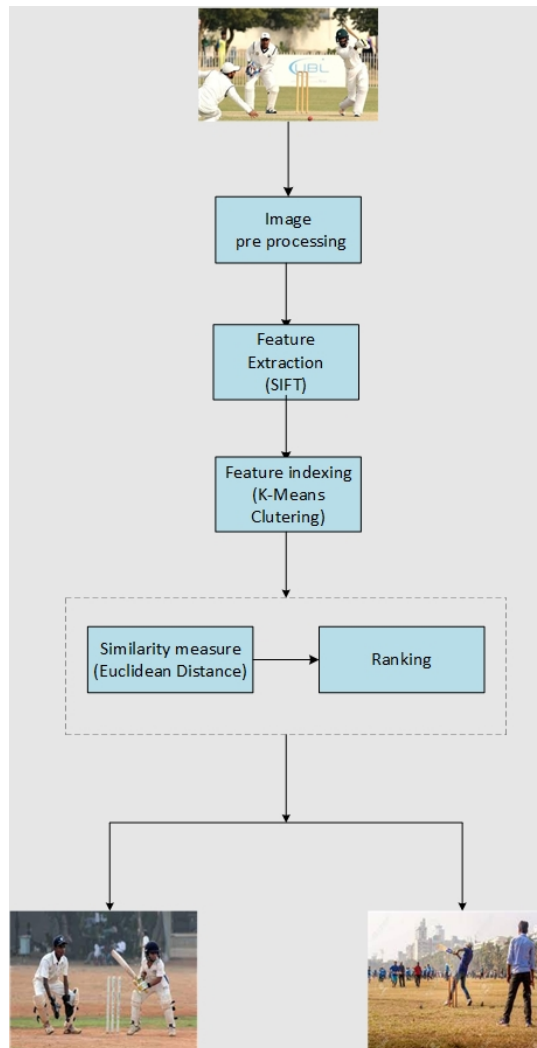


Figure 5.3: Image Retrieval subsystem Architecture Diagram

The figure above shows the subsystem architecture of the image retrieval module of the system

5.4 Architectural Strategies

The architectural strategies that we will be using for gesture decode system are given below:

5.4.1 Programming Languages

The project will leverage Python programming language as it has many built in libraries such as NLPAug which can be used for adversarial attacks, adversarial training, and data augmentation in NLP. Python's inclusion is driven by its robust libraries for data handling and widespread use in machine learning models. For a clean and user-friendly interface with high-level and swift interactions, we opt for JavaScript. The Django framework will structure the front end. The choice of JavaScript is rooted in its powerful frameworks, libraries, and component-based design structure, offering a superior approach to web application development.

5.4.2 Reuse of existing software components

The system will employ the pre-trained VGG-16 model designed to extract image features specifically for image captioning. We'll integrate the latest existing models to enhance the efficiency of image captioning and retrieval, showcasing a strategic utilization of current technologies in the system.

5.4.3 User Interface Paradigms

The interface will be straightforward and user-friendly, ensuring accessibility for all users. Every functionality will be designed with purpose, eliminating the need for additional training to use our website.

5.4.4 Database Management

XAMPP will serve as the local development environment for seamless integration, incorporating MySQL for efficient database management. This configuration is intended to store user data and generated results, streamlining the development process.

5.5 Class Diagram

The class diagram of the system is as follows:

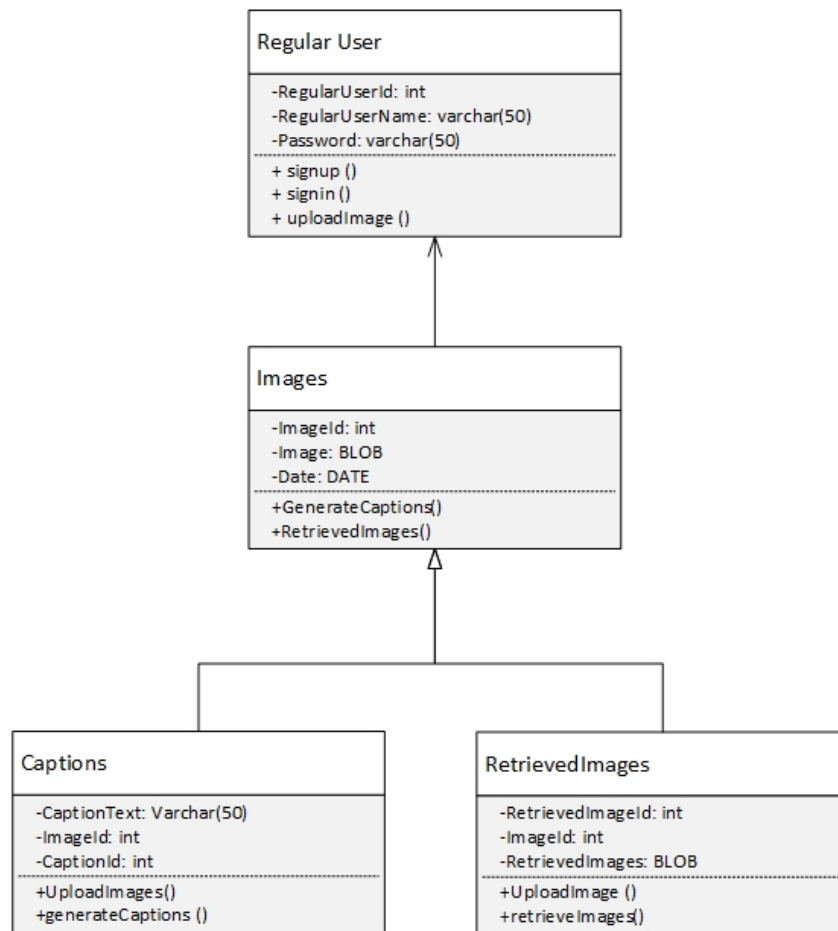


Figure 5.4: Class Diagram

The figure above shows the structure design of the system

5.6 Sequence Diagrams

The sequence diagrams of the system are as follows.

5.6.1 Sign Up

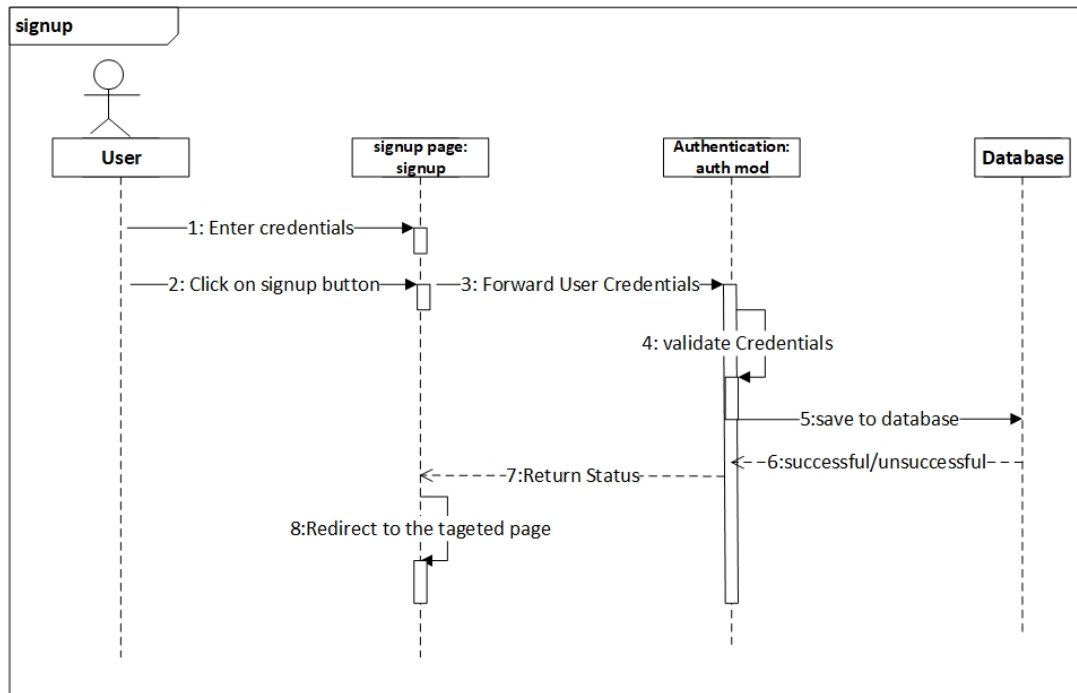


Figure 5.5: Sign up Sequence Diagram

The figure above shows that the system take required information from a new user and create an account

5.6.2 Login

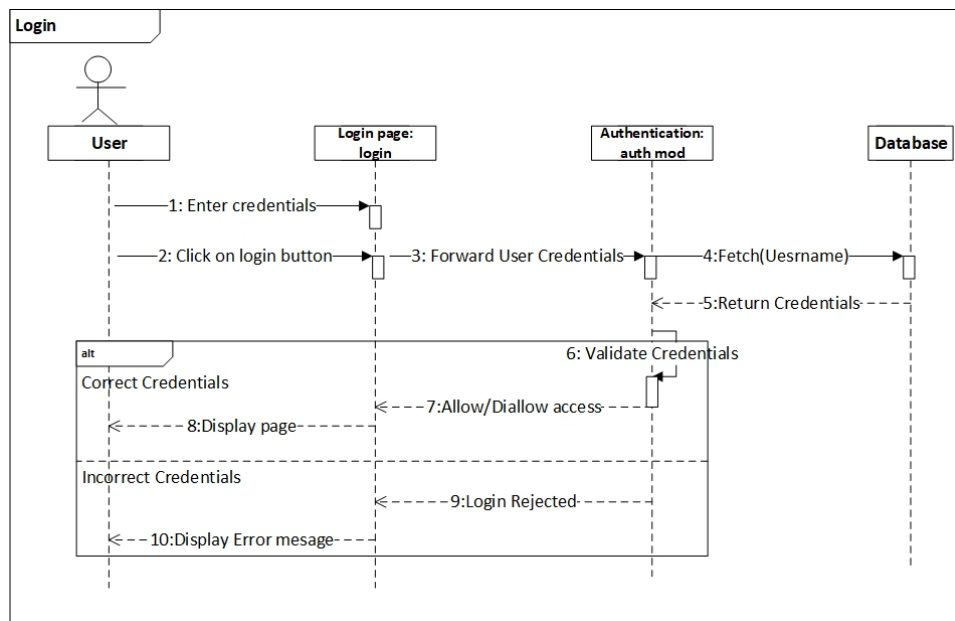


Figure 5.6: Login Sequence Diagram

The above figure shows the sequence diagram of the login functionality of the system

5.6.3 Upload Image

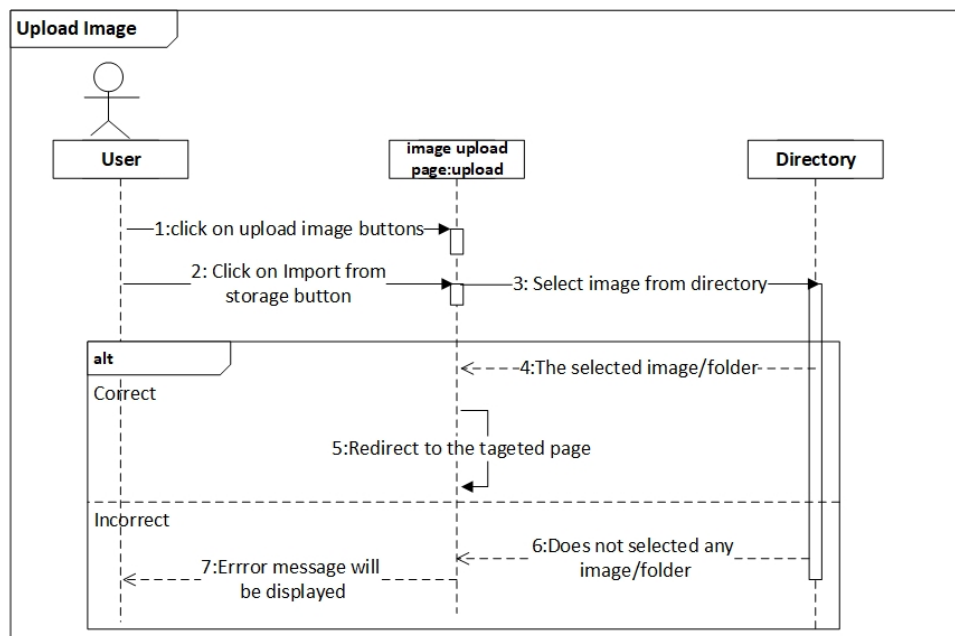


Figure 5.7: Upload Image Sequence Diagram

The above figure shows the sequence diagram of the Upload image/folder functionality of the system

5.6.4 Download Image

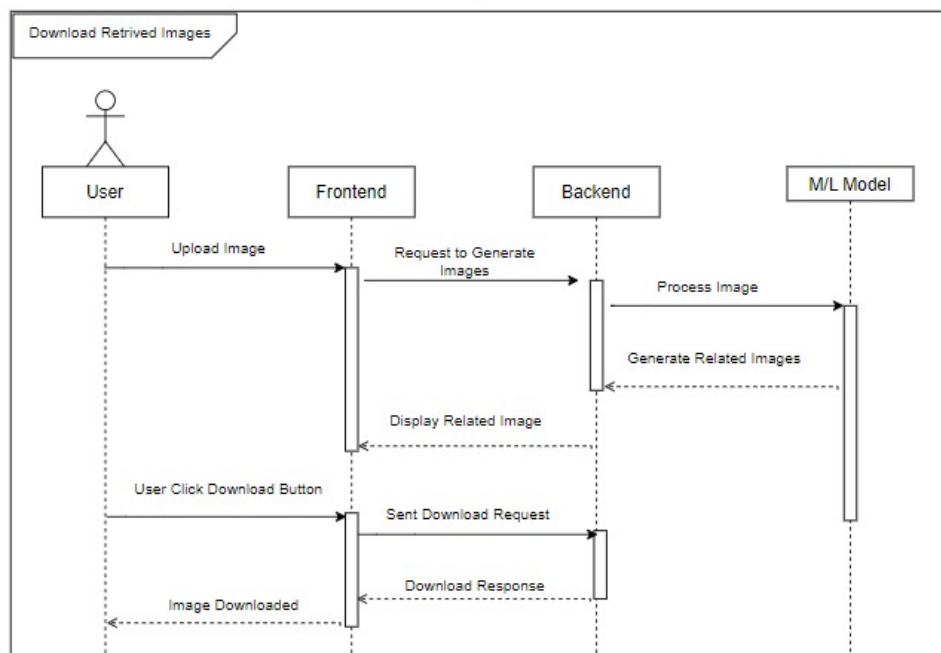


Figure 5.8: Download Retrieved Images Sequence Diagram

The above figure shows the sequence diagram to download retrieved images functionality of the system

5.6.5 History Log

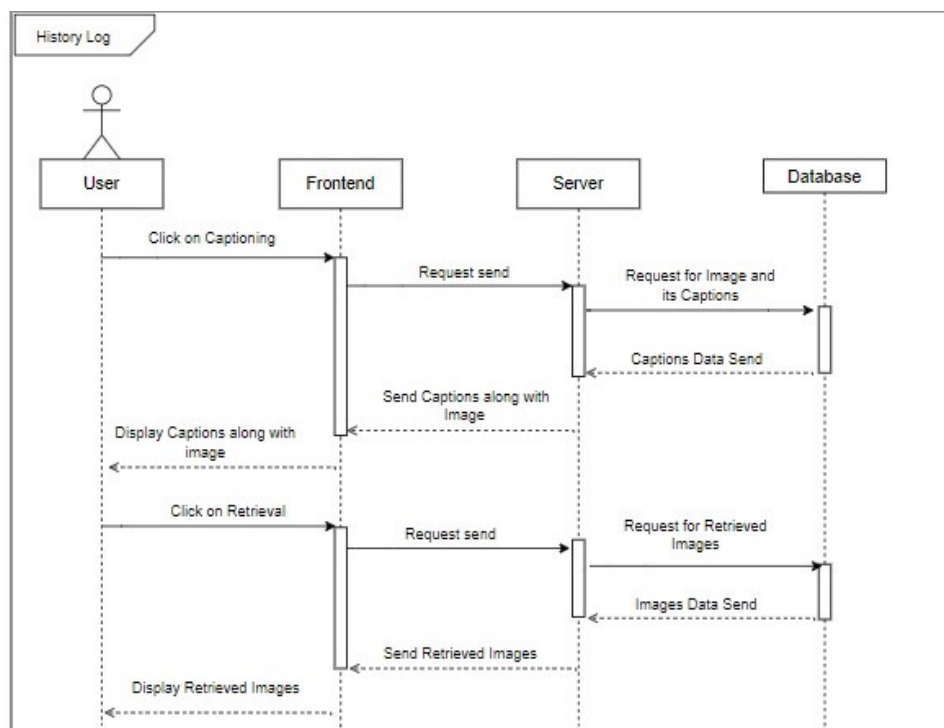


Figure 5.9: History Log Sequence Diagram

The above figure shows the sequence diagram to maintain the history log functionality of the system

5.7 Policies and Tactics

The following are the considerations for the policies that would be followed in the project.

5.7.1 Tools

The training and testing of our classification models will be conducted using Google Colaboratory and Visual Studio Code, leveraging the capabilities of both platforms. Additionally, Google Drive will serve as the designated repository for maintaining the databases crucial for training and refining our models throughout the development process.

5.7.2 Coding Guidelines

The language preferred for backend development is Python while the frontend will adopt a functional programming paradigm with Javascript and CSS for responsive design. Strict compliance to language-specific rules and standards will be upheld throughout the programming phase.

5.7.3 Extensibility

The system will be designed with extensibility in mind, anticipating the addition of new features and scalability for a growing user base. We will adopt a modular approach and will create micro-services for each component. This strategy will significantly enhance the project's integration capabilities.

5.7.4 User interface

The application will be a user friendly and modern GUI design to balance between simplicity and sophistication. We will prioritize an appealing color scheme that is both relevant and aesthetically pleasing.

5.7.5 Testing the Software

The software testing process will employ both white box and black box testing techniques to assess various modules. We will also do rigorous testing across all aspects to identify and mitigate potential errors. Finally, we will also perform the whole system test to ensure the highest quality standards are met.

5.8 Conclusion

In conclusion, this chapter provides a comprehensive system overview, outlining design considerations, architectural strategies and policies. The detailed representation through class diagrams and sequence

diagrams ensures a clear and structured visualization of the system architecture. It facilitates effective communication and understanding of the design elements and tactics employed in the development process. Together, these components contribute to a robust framework that aligns with the project's objectives and design principles.

Chapter 6 Implementation and Test Cases

This chapter describes the comprehensive implementation of our proposed idea, explaining both the extent and manner in which it has been executed. It also describes test cases to enhance understanding of the project's inputs, processing, and outputs. We aim to provide a clear understanding of the implementation process and the outcomes derived from our approach in this chapter.

6.1 Implementation

This section outlines the algorithms implemented along with the libraries utilized.

6.1.1 Web Application

We have developed the frontend for our web application using Python Django framework. We also have maintained a database for our backend purpose to store the images and the results that will be generated by our models.

6.1.2 Implementation of Text-to-Text Augmentation

NLPAug, a Python library designed for text-to-text augmentation in machine learning experiments has been used in our implementation. Specifically, we applied text-to-text augmentation as shown in Figure 6.1 to each caption of 80000 images, incorporating five distinct transformations: Word Deletion, Synonym Replacement, Embedding Swap, Inner Swap Random, and BERT-Masked LM Swap. This approach enables the generation of diverse captions for each image to have a broader range of dictionary words. The objective is to enhance the efficiency of training our image captioning models through increased textual variation.

```

Input Sentence: A child in a pink dress is climbing up a set of stairs in an entry way .

Text-to-Text Augmentation using:

Word Deletion: Child in pink dress is climbing up a set stairs an.
Synonym Replacement: A child in a pinko frock is go up up a curing of stairs in an entry way.
Embedding Swap: A Fajo in a pink Marchesa_gown be Khumbu_icefall CONCORD_NC_Casey_Mears a set of stairs in toan entry way.
Inner Swap Randomization: A child in a pink is dress up climbing set of a stairs in an entry way.
BERT-Masked LM Swap: a child in a pink dress pictured climbing in a box of boxes in an outside passage.

```

Figure 6.1: Results of Text-to-Text Augmentation

The above figure shows the captions generated using NLPAug python library

6.1.3 Implementation of Text-to-Image Augmentation

We have implemented the text-to-image augmentation by performing the three main components of stable diffusion on the flickr8K dataset. We used a combination of Hugging Face's transformers library and a library named diffusers to initialize and utilize several pre-trained models. First, an AutoencoderKL model was loaded from the "CompVis/stable-diffusion-v1-4" repository with a specific subfolder "vae" that served as a variational autoencoder (VAE) for decoding latents into image space. Next, a CLIP-Tokenizier and a CLIPTextModel were loaded from the "openai/clip-vit-large-patch14" repository that facilitated tokenization and text encoding for the CLIP model. Finally, a UNet2DConditionModel, a 2D U-Net model designed for generating latents was loaded from the same "CompVis/stable-diffusion-v1-4" repository, with a specific subfolder "unet". The autoencoder compressed images into latent codes then a text interpreter converted sentences into coded messages and a UNet model was used to create new images based on the text. The autoencoder then translated the artistic creation back into pixelated form that resulted in an image inspired by the original words. Together, these models laid the groundwork for text-to-image augmentation as shown in Figure 6.2.

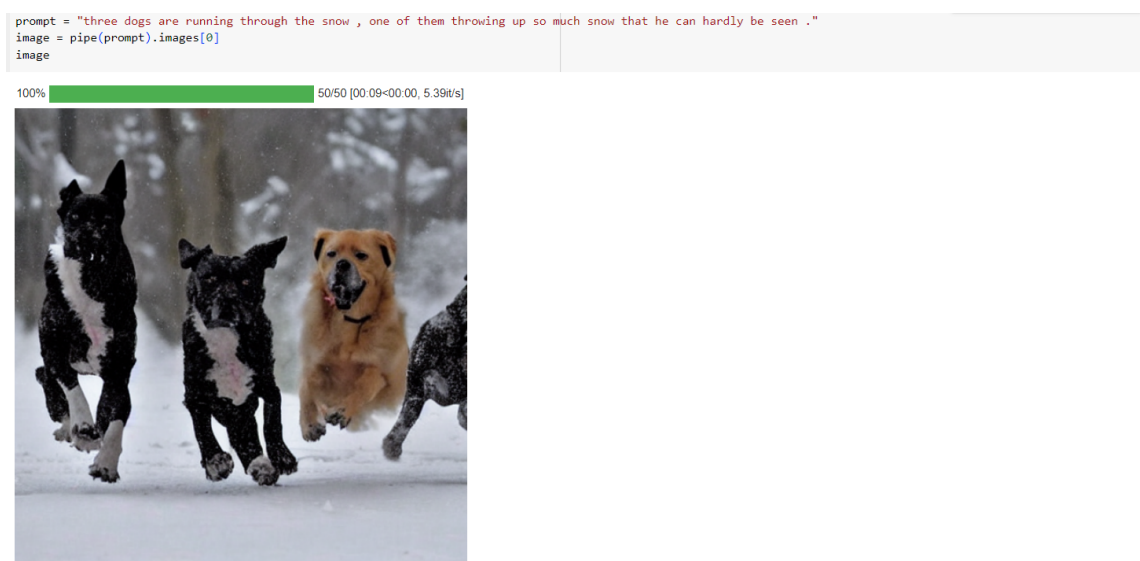


Figure 6.2: Text-to-Image Augmentation

The above figure shows the image that is generated using stable diffusion model for the caption given.

6.1.4 Implementation of Data Pre Processing

We have implemented the pre processing of both the images and the captions in the image captioning.

6.1.4.1 Captions Pre Processing

In pre processing of captions, we have performed the following steps on the dataset:

- **Tokenization:** We used 'Tokenizer()' function that is also associated with Keras and TensorFlow in Python. The words in a caption are tokenized and are transformed into a one-hot representation as shown in Figure 6.3. Subsequently, these encodings are then passed to the embeddings layer to generate word embeddings. This step was crucial for building a vocabulary.
- **Building Vocabulary:** We created a vocabulary by collecting all unique words from the tokenized captions using function `texts_to_sequences`. We have assigned a unique index to each word in the vocabulary. We also performed the padding process which ensures that all tokenized caption sequences have the same length by padding shorter sequences with a special padding token. This step was essential for converting words into numerical indices that can be fed into the model.

```
{'a': 1, 'in': 2, 'of': 3, 'child': 4, 'pink': 5, 'dress': 6, 'is': 7, 'climbing': 8, 'up': 9, 'set': 10, 'stairs': 11, 'an': 12, 'entry': 13, 'way': 14, 'grass': 15, 'plays': 16, 'with': 17, 'fingerprints': 18, 'in front of': 19, 'a white canvas': 20, 'with a rainbow': 21, 'on it': 22}

Sequence 1 in x
Input: A child in a pink dress is climbing up a set of stairs in an entry way .
Output: [1, 4, 2, 1, 5, 6, 7, 8, 9, 1, 10, 3, 11, 2, 12, 13, 14]

Sequence 2 in x
Input: A child in a pink dress is climbing up a set of stairs in an entry way .
Output: [1, 4, 2, 1, 5, 6, 7, 8, 9, 1, 10, 3, 11, 2, 12, 13, 14]

Sequence 3 in x
Input: A small girl in the grass plays with fingerprints in front of a white canvas with a rainbow on it
Output: [1, 16, 17, 2, 18, 19, 20, 15, 21, 2, 22, 3, 1, 23, 24, 15, 1, 25, 26, 27]
```

Figure 6.3: Tokenization of captions

The above figure shows the unique words and their indexes

6.1.4.2 Images Pre Processing

In image pre processing, we have performed the following steps on the dataset:

- **Image Loading:** We used "img_to_array" function that is typically associated with image processing libraries like Keras and TensorFlow in Python. It is a utility function provided by these libraries to convert images into numerical arrays, which can be used as input for machine learning models. We have resized all the images to a uniform size. This step was necessary because deep learning models require inputs of consistent dimensions.
- **Image Normalization:** We have normalized the pixel values in the images to a specific range between 0 and 1 as this helps the model converge faster during training.
- **Feature Extraction:** We also have performed feature extraction using pre-trained model VGG-16 to extract the features of images. During feature extraction, our images are passed through the convolutional layers of VGG-16. Each convolutional layer applies 3x3 filters to the input feature maps to extract local features and activating them with a non linear function (ReLU). The model is loaded and restructured to remove the fully connected layers, retaining only the layers necessary for feature extraction

6.1.5 Implementation of Image Captioning

We have implemented image captioning in following steps.

6.1.5.1 Encoder and Decoder Models

An encoder-decoder architecture is employed where the encoder extracts features from images using VGG16 and the decoder generates captions using Bi-LSTM. The encoder model includes layers for processing image features obtained from VGG16. The decoder model incorporates an embedding layer, Bi-LSTM layers and fully connected layers for generating captions.

6.1.5.2 Beam Search for Caption Generation

We have implemented a beam search with temperature sampling which is a method for generating diverse captions. The function predicts multiple captions for a given image and then selects two captions that are both diverse and relevant as shown in Figure 6.4. It accomplishes this by computing a diversity score between pairs of captions and selecting the captions with the highest diversity score. This approach ensures that the generated captions are not only diverse but also relevant to the content of the image.



Generated Caption 1 : Two dogs are playing in the grass.

Generated Caption 2 : Two dogs are playing tug of war with toy in grassy space.

Figure 6.4: Results of captioning

The above figure shows the input image and the generated captions

6.1.5.3 Evaluation

The trained model is evaluated using BLEU score, which measure the similarity between the predicted captions and the reference captions in the test dataset. We trained models on four versions of flicker8K

dataset and subsequently evaluated their performance using BLEU scores. The results of these evaluations are shown in Figure 6.5.

Augmentation method	Captioning Model	BLEU Score
No Augmentation	CNN-Bi LSTM	34.45
Image only	CNN-Bi LSTM	38.79
Text only	CNN-Bi LSTM	39.42
Text and Image	CNN-Bi LSTM	42.41

Figure 6.5: Evaluation table of models

The above figure shows the evaluation results of the models trained on four different datasets

6.1.6 Implementation of Content-Based Image Retrieval

We have implemented Content-Based image retrieval in following steps.

6.1.6.1 Convolutional Autoencoder

We have implemented a convolutional autoencoder model which is capable of encoding and decoding images. The encoder part consists of several convolutional layers followed by ReLU activation functions and max pooling operations gradually reducing the spatial dimensions of the input image while increasing the number of channels. The decoder section mirrors the encoder using transpose convolutional layers to upsample the encoded features back to the original image dimensions. The activation functions are ReLU for all layers except the last one where it utilizes a hyperbolic tangent (Tanh) activation to ensure output pixel values are within the range $[-1, 1]$. The forward method defines the flow of data through the encoder and decoder that transforms input images into encoded representations and then reconstructing them back to their original form.

6.1.6.2 Feature Extraction Using Convolutional Autoencoder and SIFT

Then we performed feature extraction using a pre-trained convolutional autoencoder model and SIFT by converting each image to a tensor and forwarding it through the encoder part of the model and storing the output latent features.

6.1.6.3 Search Nearest Images to Query Image

Then we find the closest matches to a query feature vector from a collection of feature vectors stored in an index. We use K-Means clustering and Euclidean distance between the query vector and each vector in the index to find the closest matches. Finally we return the top results, with the option to limit the number of results returned as shown in Figure 6.6.



Figure 6.6: Results of CBIR

The above figure shows the input image and the similar images computed by our model

6.1.6.4 Evaluation:

We used SSIM, which stands for Structural Similarity Index Measure, to quantify the similarity between the input image and the images generated by our model. We evaluated our model on unseen test set and the results were 74% accurate.

6.2 Test case Design and description

Below, you'll find elaboration on all the test cases conducted during the system's testing phase.

6.2.1 Sign Up Test Case

Table 6.1: Sign Up Test Case

The table below provides a comprehensive test case description for sign up.

User sign up			
Test Case ID:	1	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	7 April 2024	Use Case Reference:	4.6.1
Revision History:	None		
Objective:	To check whether the user can register using their email address.		
Product/Ver/Module:	Authentication Module		
Environment:	The website is up and running		
Assumptions:	Sign up button is visible to user		
Pre-Requisite:	The user has a valid email address and password.		
Step No.	Execution description	Procedure result	
1	The User clicks on Sign up button.	The "Signup" page is displayed.	
2	User enters username and password and clicks Sign up button.	The System registers that user as a valid user.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.2 Login Test Case

Table 6.2: Login Test Case

The table below provides a comprehensive test case description for login.

User Login			
Test Case ID:	2	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	7 April 2024	Use Case Reference:	4.6.2
Revision History:	None		
Objective:	To check if user is able to successfully login to the application		
Product/Ver/ Module:	Authentication Module		
Environment:	The website is up and running		
Assumptions:	Login button is visible to user		
Pre-Requisite:	The user's information is already recorded in the database, and he/she is a registered user.		
Step No.	Execution description	Procedure result	
1	The User clicks on login button.	The "Login" page is displayed.	
2	User enters username and password and clicks login.	The "Home page" is displayed.	
Comments: The test case has passed indicating that our system functions as intended.			
Passed Failed Not Executed			

6.2.3 Sign Out Test Case

Table 6.3: Sign Out Test Case

The table below provides a comprehensive test case description for sign out.

User sign out/logout			
Test Case ID:	3	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	7 April 2024	Use Case Reference:	4.6.3
Revision History:	None		
Objective:	To check whether the user’s information is removed from the database		
Product/Ver/Module:	Authentication Module		
Environment:	The website is up and running		
Assumptions:	Sign out button is visible to user		
Pre-Requisite:	The user has already logged into the system		
Step No.	Execution description	Procedure result	
1	The User clicks on Sign out button.	System will remove all information of user from Database.	
2	The User clicks on Sign out button.	”Signup” page will be displayed.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.4 Home Screen Test Case

Table 6.4: Home Screen Test Case

The table below provides a comprehensive test case description for Home Screen.

Home Screen			
Test Case ID:	4	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	7 April 2024	Use Case Reference:	4.6.4
Revision History:	None		
Objective:	To check whether the buttons of image captioning, retrieval and history are working.		
Product/Ver/Module:	Home Screen		
Environment:	The website is up and running		
Assumptions:	All buttons on the home page are visible to user.		
Pre-Requisite:	The user has already logged into the system		
Step No.	Execution description	Procedure result	
1	The user clicks on Captioning/Retrieval button.	System will display "Upload image" page to the user.	
2	The user clicks on History button.	System will display history page to the user.	
Comments: The test case has passed indicating that our system functions as intended.			
Passed Failed Not Executed			

6.2.5 Image Upload for Captioning Test Case

Table 6.5: Image Upload for Captioning Test Case

The table below provides a comprehensive test case description pertaining to the image upload for image captioning process.

Image Uploading for image captioning			
Test Case ID:	5	QA Test Engineer:	Mahnoor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	14 April 2024	Use Case Reference:	4.6.5
Revision History:	None		
Objective:	To check if a user can upload images		
Product/Ver/Module:	Image Captioning Module		
Environment:	The website is up and running		
Assumptions:	Upload Image button is visible to the user		
Pre-Requisite:	The user has already logged into their account.		
Step No.	Execution description	Procedure result	
1	User clicks the "Upload Image" button.	The "Upload Image" page is displayed.	
2	User selects the images	The system uploads the images	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.6 Correct File Format Test Case

Table 6.6: Correct File Format Test Case

The table below provides a comprehensive test case description pertaining to notify the user to upload the correct format file .

Correct File Format			
Test Case ID:	6	QA Test Engineer:	Mahnoor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	14 April 2024	Use Case Reference:	4.6.5
Revision History:	None		
Objective:	To check if a user is uploading correct format file		
Product/Ver/Module:	Image Captioning Module		
Environment:	The website is up and running		
Assumptions:	System has the ability to detect the format of the uploaded file		
Pre-Requisite:	The user has already uploaded the file as an input.		
Step No.	Execution description	Procedure result	
1	User uploads the wrong format file.	The system shows the error "Incorrect File Format".	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.7 Results of Captioning Test Case

Table 6.7: Results of Captioning Test Case

The table below provides a comprehensive test case description pertaining to generate the captions.

Results of Captioning			
Test Case ID:	7	QA Test Engineer:	Mahnoor
Test case Version:	1.0	Reviewed By:	Rida Noor
Test Date:	16 April 2024	Use Case Reference:	4.6.6
Revision History:	None		
Objective:	To check if model is generating correct captions		
Product/Ver/Module:	Image Captioning Module		
Environment:	The website is up and running		
Assumptions:	User can upload image of any kind.		
Pre-Requisite:	The user has uploaded image files.		
Step No.	Execution description	Procedure result	
1	User selects the image files.	The system displays the related captions.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.8 Switch Between Results Test Case

Table 6.8: Switch Between Results Test Case

The table below provides a comprehensive test case description pertaining to check if a user can switch between the results.

Switch between results			
Test Case ID:	8	QA Test Engineer:	Mahnoor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	16 April 2024	Use Case Reference:	4.6.6
Revision History:	None		
Objective:	To check if a user can switch between the results		
Product/Ver/Module:	Image Captioning Module		
Environment:	The website is up and running		
Assumptions:	"Next" button and "previous" button is visible to the user		
Pre-Requisite:	The System has generated the captions of an image/images.		
Step No.	Execution description	Procedure result	
1	User clicks the "Next" button.	The system shows the next image captions.	
2	User clicks the "Previous" button.	The system shows the previous image captions.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.9 Download Captions Test Case

Table 6.9: Download Captions Test Case

The table below provides a comprehensive test case description pertaining to download the captions.

Download captions			
Test Case ID:	9	QA Test Engineer:	Mahnoor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	16 April 2024	Use Case Reference:	4.6.6
Revision History:	None		
Objective:	To check if a user can download the captions		
Product/Ver/Module:	Image Captioning Module		
Environment:	The website is up and running		
Assumptions:	Download button is visible to the user		
Pre-Requisite:	The user has already uploaded the image/images as an input.		
Step No.	Execution description	Procedure result	
1	User clicks the "Download" button.	System downloads the Captions text file in downloads.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.10 Image Upload for Retrieval Test Case

Table 6.10: Image Upload for Retrieval Test Case

The table below provides a comprehensive test case description pertaining to the image upload for retrieval process.

Image Uploading for Retrieval			
Test Case ID:	10	QA Test Engineer:	Zeemal Nadeem
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	7 April 2024	Use Case Reference:	4.6.7
Revision History:	None		
Objective:	To check if a user can upload an image		
Product/Ver/Module:	Image Retrieval Module		
Environment:	The website is up and running.		
Assumptions:	"Upload Image" button is visible to the user.		
Pre-Requisite:	The user has already logged into their account.		
Step No.	Execution description	Procedure result	
1	User clicks the "Upload Image" button.	The "Upload Image" page is displayed.	
2	User selects the image.	The system uploads the image.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.11 Notification for Wrong File Uploads Test Case

Table 6.11: Notification for Wrong File Uploads Test Case

The table below details test case for image uploads, with notifications for users if the input file isn't image.

Notification for Wrong File Uploads			
Test Case ID:	11	QA Test Engineer:	Zeemal Nadeem
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	11 April 2024	Use Case Reference:	4.6.7
Revision History:	None		
Objective:	To notify user that the uploaded file is not an image.		
Product/Ver/Module:	Image Retrieval Module		
Environment:	The website is up and running.		
Assumptions:	System has the capability to accurately detect file types, distinguishing between images and other file formats.		
Pre-Requisite:	The user has already uploaded a file as input.		
Step No.	Execution description	Procedure result	
1	User selects a file other than an image.	A notification "Incorrect File Format" will be displayed.	
Comments: The test case has passed indicating that our system functions as intended.			
Passed Failed Not Executed			

6.2.12 Results of Retrieval Test Case

Table 6.12: Results of Retrieval Test Case

The following table provides a comprehensive test case pertaining to the results of image retrieval models.

Results of Retrieval			
Reference			
Test Case ID:	12	QA Test Engineer:	Zeemal Nadeem
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	25 May 2024	Use Case Reference:	4.6.8
Revision History:	None		
Objective:	To check if the system displays similar images.		
Product/Ver/Module:	Image Retrieval Module		
Environment:	The website is up and running.		
Assumptions:	Input images must be similar to the images on which model is trained on.		
Pre-Requisite:	The system has given an image file as input.		
Step No.	Execution description	Procedure result	
1	User uploads an image file	System displays the similar images	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.13 Switching Retrieval Results Test Case

Table 6.13: Switching Retrieval Results Test Case

The following table provides a comprehensive breakdown of test case pertaining to the switching process between output images of retrieval.

Switching Retrieval Results			
Test Case ID:	13	QA Test Engineer:	Zeemal Nadeem
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	25 April 2024	Use Case Reference:	4.6.8
Revision History:	None		
Objective:	To check if a user can switch between image retrieval results.		
Product/Ver/Module:	Image Retrieval Module		
Environment:	The website is up and running.		
Assumptions:	"Next" and "previous" buttons are visible to the user.		
Pre-Requisite:	The system has generated multiple images.		
Step No.	Execution description	Procedure result	
1	User clicks on the "Next" button.	System displays the next image.	
2	User clicks on the "Previous" button.	System displays the previous image.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.14 Downloading Retrieval Results Test Case

Table 6.14: Downloading Retrieval Results Test Case

The table below offers a detailed breakdown of test case relevant to downloading output images for retrieval.

Downloading Retrieval Results			
Test Case ID:	14	QA Test Engineer:	Zeemal Nadeem
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	25 April 2024	Use Case Reference:	4.6.8
Revision History:	None		
Objective:	To check if a user can download Image Retrieval results for their input image.		
Product/Ver/Module:	Image Retrieval Module		
Environment:	The website is up and running.		
Assumptions:	"Download Image" button is visible to the user.		
Pre-Requisite:	The user has already uploaded an image as input.		
Step No.	Execution description	Procedure result	
1	User clicks on the "Download Image" button.	System saves the output images to the Downloads directory.	
2	User selects image from local directory.	The system uploads the image.	
Comments: The test case has passed indicating that our system functions as intended.			
Passed Failed Not Executed			

6.2.15 Captioning History Test Case

Table 6.15: Captioning History Test Case

The table below offers a description of captioning history.

Captioning History			
Test Case ID:	15	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	25 April 2024	Use Case Reference:	4.6.9
Revision History:	None		
Objective:	To check if user’s captions history is maintained in Database.		
Product/Ver/Module:	Image Captioning History Module		
Environment:	The website is up and running.		
Assumptions:	”View History” button is visible to the user.		
Pre-Requisite:	The user has already uploaded an image as input to generate its captions.		
Step No.	Execution description	Procedure result	
1	User clicks on the ”View History” button.	System displays history page.	
2	User selects the image captioning from the two given options.	The system displays all images uploaded for image captioning.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.16 Retrieval History Test Case

Table 6.16: Retrieval History Test Case

The table below offers a description of retrieval history.

Retrieval History			
Test Case ID:	16	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	25 April 2024	Use Case Reference:	4.6.9
Revision History:	None		
Objective:	To check if user’s retrieval history is maintained in database.		
Product/Ver/Module:	Image Retrieval History Module		
Environment:	The website is up and running.		
Assumptions:	”View History” button is visible to the user.		
Pre-Requisite:	The user has already uploaded an image as input to retrieve related images.		
Step No.	Execution description	Procedure result	
1	User clicks on the ”View History” button.	System displays ”History” page.	
2	User selects the image Retrieval from the two given options.	The system displays all images uploaded for image retrieval.	
Comments: The test case has passed indicating that our system functions as intended.			
Passed Failed Not Executed			

6.2.17 Managing Seperate History Test Case

Table 6.17: Managing Seperate History Test Case

The following table provides a comprehensive test case related to history maintenance of captioning and retrieval separately.

Seperate History			
Test Case ID:	17	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	25 April 2024	Use Case Reference:	4.6.9
Revision History:	None		
Objective:	To check if user’s retrieval and captioning history is maintained separately.		
Product/Ver/Module:	History Module		
Environment:	The website is up and running.		
Assumptions:	”View History” button is visible to the user.		
Pre-Requisite:	The user has already uploaded images for both captioning and retrieval.		
Step No.	Execution description	Procedure result	
1	User clicks on the ”View History” button.	System displays ”History” page.	
2	User can choose any option between Captioning and retrieval to see the history.	The system displays relevant page based upon user’s selection.	
Comments: The test case has passed indicating that our system functions as intended.			
☒ Passed ☐ Failed ☐ Not Executed			

6.2.18 Maintaining Date Stamp Test Case

Table 6.18: Maintaining Date Stamp Test Case

The following table provides a comprehensive test case related to image Upload Log with Date Stamp

Image Upload Log with Date Stamp			
Test Case ID:	18	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Zeemal Nadeem
Test Date:	25 April 2024	Use Case Reference:	4.6.9
Revision History:	None		
Objective:	To keep track of the date on which a particular image was upload on the system		
Product/Ver/Module:	History Module		
Environment:	The website is up and running.		
Assumptions:	"View History" button is visible to the user.		
Pre-Requisite:	The user has already uploaded images for both captioning and retrieval.		
Step No.	Execution description	Procedure result	
1	User can choose any option between Captioning and retrieval to see the history.	The system displays each image along with its uploaded date.	
Comments: The test case has passed indicating that our system functions as intended.			
Passed Failed Not Executed			

6.2.19 History Deletion Test Case

Table 6.19: History Deletion Test Case

The following table provides a comprehensive test case related to image delation from the Database

History Deletion			
Test Case ID:	19	QA Test Engineer:	Rida Noor
Test case Version:	1.0	Reviewed By:	Mahnoor
Test Date:	25 April 2024	Use Case Reference:	4.6.10
Revision History:	None		
Objective:	To remove/delete any image from the database.		
Product/Ver/Module:	Image Deletion Module		
Environment:	The website is up and running.		
Assumptions:	”View History” button is visible to the user.		
Pre-Requisite:	The user has already uploaded images for both captioning and retrieval.		
Step No.	Execution description	Procedure result	
1	User selects an image.	The system displays an option to delete.	
Comments: The test case has passed indicating that our system functions as intended.			
	Passed	Failed	Not Executed

6.3 Test Metrics

Following are the test metrics for functional and non functional requirements.

6.3.1 Functional Test Metrics

Table 6.20: Functional test case metric

The following table shows the test case metric for testing the functional requirements of the system.

Metric	Value
Number of Test Cases	19
Number of Test Cases Passed	19
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	0

6.3.2 Non Functional Test Metrics

Table 6.21: Non functional test case metric

The following table shows the test case metric for testing the non functional requirements of the system.

Metric	Value
Number of Test Cases	5
Number of Test Cases Passed	5
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	0

6.4 Conclusion

In conclusion this chapter offers a thorough overview of the implementation of our proposed idea detailing its execution and the test cases employed to clarify the projects inputs, processing and outputs. Our objective is to provide a clear comprehension of the implementation process and the resultant outcomes achieved through our approach.

Chapter 7 User Manual

This chapter covers the instructions to use the application, describes its main features, and outlines the workflow for users.

7.1 Sign up

- Locate the "Sign up" button on the website.
- Click on the "Sign up" button to access the registration form.
- Fill in the required information, including your name , password and confirm password. Make sure your password is secure, using a mix of numeric and alphabets.
- Once you've filled out the form, click "Sign up" to initiate the account creation process.

7.2 Login

- Click on the "login" button to open the login form.
- Fill the information with your username and password.
- Then to submit the form click on the "login" button which will redirect you to the home page.

7.3 Logout

- After successfully login you can also logout from the application.
- Click "Logout" button to logout which will redirect you to the Sign up page..

7.4 Captioning

- After successfully login system will redirect you to the home page.
- Find the "Captioning" button there.
- Click on the "Captioning" button. Which will redirect you to the upload image page.
- Click on the "Upload" button and upload the image/images from your local directories.
- System will show you the output captions of the image/images.

7.4.1 Switch Between The Captioning Results

- After the system has generated the captions you can switch between the results.
- Click on the "Next" button to go to the next image caption.
- Click "previous" button to go to the previous image caption.

7.4.2 Download Captioning Results

- After the system has generated the captions you can download them.
- Click on the "Download" button at the bottom of the captioning results.
- After this system will download the text file of the output captions to the downloads.

7.5 Retrieval

- After Successfully login system will redirect you to "Home" page.
- Find "Retrieval" button there.
- Clicks on "Retrieval" button which will redirect you to the upload image page.
- Click "Upload" button to upload an image.
- System will show the output retrieved images of input image.

7.5.1 Switch Between The Retrieval Results

- After the system has generated the related images you can switch between the results.
- Click on the "Next" button to go to the next image.
- Click "previous" button to go to the previous image.

7.5.2 Download Retrieval Results

- After the system has retrieved the related images you can download them.
- Click on the "Download" button at the bottom of the each resultant image.
- After this system will download the respective image file to the downloads.

7.6 History

- After Successfully login system will redirect you to "Home" page.
- Find "View History" button there.
- Clicks on "View History" button which will redirect you to the "History" page.

7.6.1 Captions History

- Find navigation bar present on left side of page.
- Click "Captioning" to see all the input image that you have uploaded for captioning.

7.6.2 Retrieval History

- Find navigation bar present on left side of page.
- Click "Retrieval" to see all the input image that you have uploaded for retrieval.

7.6.3 Delete History

- Find "delete" button on the "History" page.
- Click "delete" button to delete the respective image from user history.

7.7 Conclusion

The User Manual provides a comprehensive guide on key features of the software platform, detailing the processes for user sign up, login, captioning, image retrieval, and viewing user history. Together, these sections offer a detailed step by step guide for users, ensuring they can navigate and utilize the platform's features effectively. By following these instructions, users can sign up, log in, generate captions, retrieve related images, and manage their history efficiently, demonstrating the platform's versatility and user friendliness.

Chapter 8 Conclusion and Future Work

The following is the conclusion and the future work that would be followed in the project.

8.1 Conclusion

In conclusion, the project aims to develop a software system focused on image captioning and retrieval with an emphasis on successful execution of defined objectives and addressing associated challenges. The preferred approaches include Stable Diffusion for text-to-image augmentation and Transformer models for image captioning along with SIFT for image retrieval. These methodologies contribute to enhanced image processing capabilities, improved caption generation and efficient retrieval that will collectively enrich the overall performance and utility of the software system.

Anticipated challenges lie in the optimization of machine learning algorithms and in the seamless integration between captioning and retrieval components but the project is aimed to the creation of a cohesive and functional system.

8.2 Future Work

We will focus on the following modules in future.

8.2.1 Results Improvement for CBIR

Our focus lies in enhancing the outcomes of image retrieval through the exploration of IGA.

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